Tech pack: CORRIE F rev 3 (door D2) - 5200 cu in hunting pack bag

Revision 3, door D2, crown kit M shown (S / L cut lists in cutting_plan.csv). Issued 20261113. Generated by designs/pack_5200/mfg_corrie_f_rev3.py from the pattern module; patterns, DXF and markers in ../patterns/. Units mm, N, g (imperial alongside).

Item	Value
Pieces	80 piece types; 77 fabric / sheet pieces; 78 webbing / cord / elastic
Ledger reconciliation	116 rows, 0 unmatched, 0 pieces without a row; hardware rows unmapped: 0
Mass from pattern areas	570.9 g vs ledger 571.5 g (-0.10 %, limit 3 %); bag ledger base 1160.8 g / as hunted 1442.6 g / meat trip 1638.6 g (team B basis); primary fabrics +95.5 g
Thread	18.84 g in the bag from the op lengths (no cone waste); ledger books 18.78 g; purchase quantities in the BOM include 15% cone waste
Operations	88 ops; beginner 74.8 h; factory SAM 509 min
One bag to buy (prototype)	$510 (1-bag markers, shop increments; costing section)
Mass targets, PRIMARY fabrics (QC-14)	base 1226 g / as hunted 1530 g / meat trip 1735 g, +/- 3 %
Pre-cut reviews (BL-040 / 041 / 043)	third-review items open: NOT ready to cut
Cost a bag (production)	materials $381 primary / $381 alternate; + CMT: US $922, Asia $457 (ASSUMPTIONS)
Load points	32; lowest SF 1.36 (B-25 bridle toggle tab); every load point is released by a proof test (section 8)
HOLD	none
Pattern flags	none
Volatile (re-run on a pattern change)	pattern: B-22 x; pattern: F-01; pattern: FP width; pattern: HE / B-18; pattern: ledger; pattern: ledger (US-chan)
Fabric sets	PRIMARY = today's fabrics (build this). ALTERNATE = team B set (ULTRAGrid body, 70D collar, ULTRA 200X floor) only after gates ALT-1..3 and FS-O5.
Owner rules	No hook-and-loop; coil zips; mostly solid colours, camo on accents only; width <= 394 mm (15.5 in); 85 L +/- 3.

Contents: 1 cover; 2 equipment and stitch standards; 3 flats / piece maps; 4 BOM; 5 cutting plan; 6 operation sequence; 7 load points and sketches; 8 QC, release tests, defects; 9 packaging; 10 costing; 11 delta register; 12 assumptions.

2. Equipment and stitch standards

ID	Machine / tool
PROTO	MINIMUM PROTOTYPE MACHINE (buy this or better): a walking-foot (compound or upper-and-lower feed) lockstitch that takes the 135x17 / DPx17 needle system up to size 140/22, sews tex 135 (V-138 / No. 138) bonded nylon top and bobbin, and clears at least 6 mm under the raised foot (38 mm webbing folded in half + Hypalon + laminate is about 5 mm), AND sews a 304 zigzag up to 5 mm wide (BL-041 Y-3: the BT-S / BT-M tacks and elastic laps are ISO 304; e.g. a portable straight / zigzag walking foot). A straight-stitch-only walking foot (the WF2 class) may be used if every 304 tack is sewn as the BT-301 fallback below. A domestic or 'heavy-duty domestic' machine (130/705H needles, tex 90 max) CANNOT sew the load points as specified. The alternative, re-rating every box-X for Nm 120/19 + tex 90 and re-running MT-1 / PT-1..PT-6 on it, is an owner decision, not taken here.
BT-301	BT-301 (prototype, straight stitch only): 4 passes over the tack length (16 mm for BT-M, 10 mm for BT-S and slit ends, the full lap width for elastic laps) at 2 mm stitch, needle down and pivot at each end, 3-stitch backtack, same thread and needle as the 304 tack it replaces. UNRATED, like every 304 tack in this release (none is on a load-table path); the PT-3 neck tacks are sewn the same way as the bag's.
WF1	Single-needle compound-feed walking foot lockstitch, Juki DU-1181N (or LU-1508N). Prototype: the PROTO machine above (it also runs Nm 110 + tex 90 for the body seams).
WF2	Heavy single-needle walking foot lockstitch, large hook, Juki DNU-1541S (or LU-1561/2210 class): webbing stacks, Hypalon, tex 135 box-X. Prototype: WF1 hand-cranked through the thick spots.
BT	Computer bartacker Juki LK-1900BN (ISO 301 path / 304 zigzag, programmable, 40 x 30 mm field): every 25 mm box-X and the slit bartacks. Prototype: box-X on WF2 with the J1 template.
PS	Programmable pattern sewer Juki AMS-210EN (220 x 100 mm field) with a webbing clamp: the 38 mm box-X that do not fit the LK-1900 field (HE, T3, B-24). Prototype / small shop: WF2 with the J1 template.
ZZ	Zigzag lockstitch (ISO 304), Juki LZ-2290 class or a portable straight / zigzag walking foot: elastic laps and BT-S / BT-M tacks. Prototype without zigzag: the BT-301 fallback.
LASER	CO2 laser cutter 60-100 W, bed >= 600 x 900 mm, fume extraction: Hypalon, TPU-840D, B-12, port bindings, HDPE <= 0.8 mm (or shear + round the corners).
HK	Hot knife, blade type, on a steel / glass plate with the J8 length stop: webbing, elastic, cord.
SK	Straight knife, 5 in blade (Eastman Blue Streak II class) on a cutting table for production lays. Prototype: 45 / 60 mm rotary cutter, 600 x 900 mat, 6 x 24 in ruler.
DIE	Clicker press + steel-rule dies: only above about 500 bags a year (Hypalon tabs, B-12, keepers).
TAPE	Hot-air seam-tape machine: NOT used. Rev 3 has no taped seams (ASSUMPTION NO-TAPE: the bag is water resistant, not waterproof; contents go in the user's dry bag). Needed only for a waterproof variant.

Use	Thread (record)	Needle	SPI (mm)	Stitch
40D-100D, mesh, silpoly	bonded_nylon_tex70	Nm 90/14 DPx17, round point R / Nm 90/14 DPx17, medium ball SUK (knit mesh)	10 (2.5)	301
body seams, zips, Hypalon perimeters, B-11 sew-down	bonded_nylon_tex90	Nm 110/18 DPx17, round point R	8 (3.2)	301
load box-X	bonded_nylon_tex135	Nm 140/22 DPx17, round point R	7 (3.6)	301 path (LK-1900 / AMS-210EN programmed)
bartacks, elastic laps	bonded_nylon_tex90	Nm 110/18 DPx17, round point R	LK-1900 std	304

Round point (R) or slim set point on every coated fabric, laminate, Hypalon and webbing. Never a leather / cutting point (LR, S, DI): it cuts the reinforcement and the coating and drops seam strength. Ball point (SUK) only on the power mesh. New needle every bag (prototype) or every 8 h (production).

MINIMUM PROTOTYPE MACHINE (buy this or better): a walking-foot (compound or upper-and-lower feed) lockstitch that takes the 135x17 / DPx17 needle system up to size 140/22, sews tex 135 (V-138 / No. 138) bonded nylon top and bobbin, and clears at least 6 mm under the raised foot (38 mm webbing folded in half + Hypalon + laminate is about 5 mm), AND sews a 304 zigzag up to 5 mm wide (BL-041 Y-3: the BT-S / BT-M tacks and elastic laps are ISO 304; e.g. a portable straight / zigzag walking foot). A straight-stitch-only walking foot (the WF2 class) may be used if every 304 tack is sewn as the BT-301 fallback below. A domestic or 'heavy-duty domestic' machine (130/705H needles, tex 90 max) CANNOT sew the load points as specified. The alternative, re-rating every box-X for Nm 120/19 + tex 90 and re-running MT-1 / PT-1..PT-6 on it, is an owner decision, not taken here.

Seam efficiency: a 301 seam is taken at 80% of the weaker strip (clothsim/fabrics.yaml, attach_system.SEAM_EFF); box-X capacity = 0.5 x 1.5 x thread break / stitch length x path (attach_system.box_x, ETA_BOX 0.5 ASSUMPTION).

Code	Seam construction
SSa2	SSa plain seam, 12 mm allowance, then both allowances turned to one side and caught by a 2nd row at 6 mm (ISO 4916 1.01.01, two rows; no binding, see ASSUMPTION NO-BIND)
SSa	SSa plain seam, single row (ISO 4916 1.01.01)
SSa-ins	SSa with a webbing tab / loop inserted between the plies (ISO 4916 class 1 with insert); tab ends reach the cut edge, a backtack each side of the tab

Code	Seam construction
LSd	LSd lapped patch: piece laid flat on the panel, perimeter row(s) (ISO 4916 class 2, lapped)
EFa	EFa edge turned once and edge-stitched (ISO 4916 class 6, edge finish)
EFb	EFb edge turned twice, stitched (ISO 4916 class 6); hem round a stiffener strip
ZIP	zip set: each tape laid face down on its panel edge, sewn at 5 mm, turned and edge-stitched at 3 mm (ISO 4916 class 1, two rows per tape)
BOX	box-X: rectangle + both diagonals through the webbing stack (ISO 4915 301 path; ISO 4916 class 7 when it fixes the webbing to a panel)
BT	bartack, ISO 4915 304 zigzag lockstitch, LK-1900 standard pattern
DART	dart: fold on the dart centre, sew leg to leg, tie off at the apex (ISO 4916 1.01.01)

Jig	What
J1	Box-X templates, clear acrylic 2 mm, one per box size in the load table (section 7 / SK-1), slots for chalk on the rectangle and both diagonals; production: LK-1900 / AMS-210EN programs of the same paths
J2	Swell window corner template + clip board (8 swells, notch to notch)
J3	B-27 port gauge: 6 mm turn, 20 mm lap, facing line; a slotted ruler
J4	Webbing fold gauge: aluminium angle with stops at 15 / 30 / 45 / 50 / 60 / 100 mm
J5	LK-1900 / AMS-210EN work clamps for 25 and 38 mm webbing stacks (Hypalon + panel)
J6	Door zip board: B-02door + B-02u laid flat with the corner-tangent notches pinned to the board
J7	Lid dart template (B-29a, 4 darts, apex drill marks)
J8	Hot-knife length fence with a scale in mm and in (cut lengths in cutting_plan.csv)

3. Flats and piece maps (from the pattern pieces)

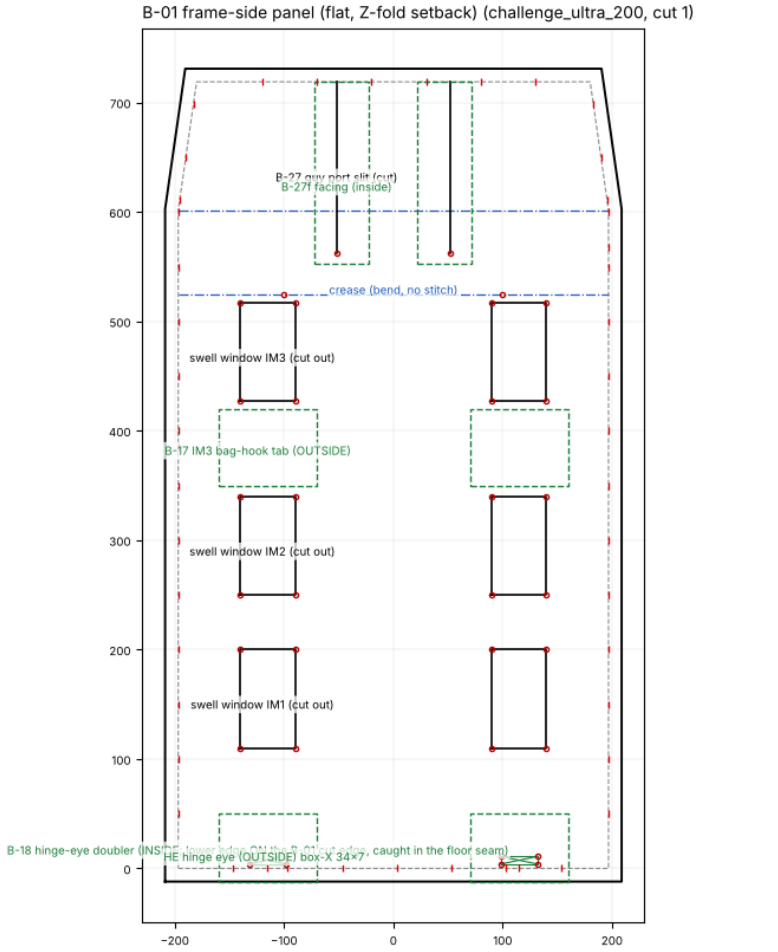


Side profile, 6-day load (45 L)

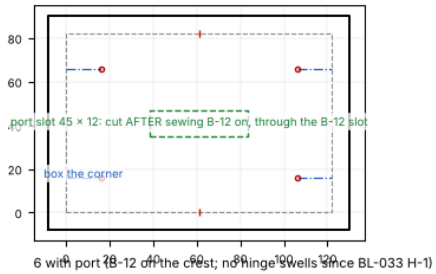
CORRIE F rev 3 (D2), H1 frame, belt B. Crown straps over the lid. Rendered 3D figure, not a photograph

Render of rev 3 (docs/screens/20261112_BL030_2.png, BL-030): side view, lid and crown straps. Colours / camo placement as drawn there.

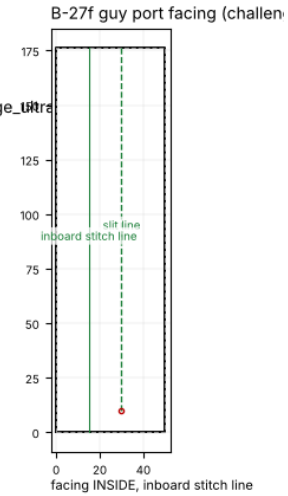
SK-3 B-01 frame-side panel: flat work in order 4.02 swells, 4.03 B-27 ports, 4.04 B-17 (outside), 4.05 B-18 + HE (inside), 4.06 I2K



6 swell windows (cut-outs), B-27 slits x +/-52 z 595-745. HE / B-18 box drawn (read from the pattern).



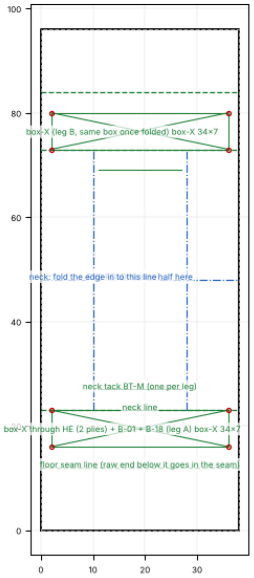
6 with port (B-12 on the crest; no hinge swells since BL-033 H-1)



facing INSIDE, inboard stitch line

SK-8 Hinge eye HE + B-18 (DECISIONS HE-ONE + HE-PIN + HE-ENV, op 4.05; PT-3, SEW-F HE and HE-FIT specimens are sewn exactly so)

(a) HE strip, flat
(38 × 96 mm)



(b) op 4.05 order (PT-3 / SEW-F HE specimens: the same)

0 CALIPER one ply of the roll: <= 1.6 mm
(3 plies must pass the 6.5 mm gap
under the pin), else change webbing.

1 NECK (flat strip): fold both long edges 10 mm in between the neck lines (18 mm wide); BT-M tack across each leg at the tack lines.

2 B-18 ROWS FIRST: B-18 inside B-01, lower edge on the B-01 cut edge; sew its 3 / 9 mm rows (top + both sides) through B-18 + B-01 ONLY.

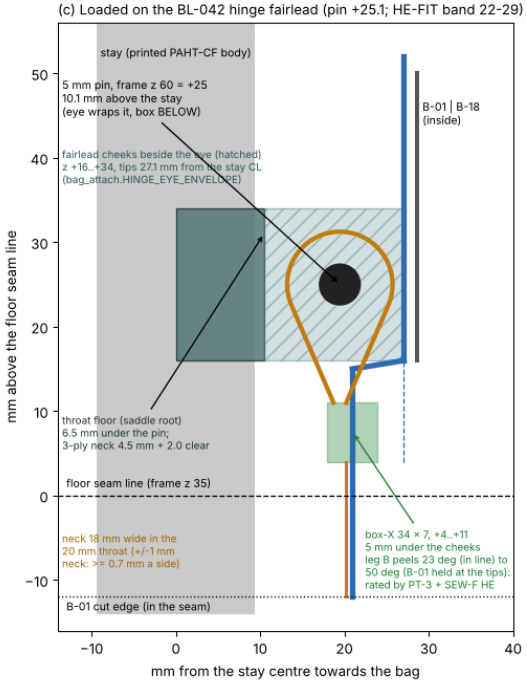
3 PLACE HE: fold in half on EYE FOLD (neck folds INSIDE), lay it OUTSIDE B-01 on the CL notch, eye UP, raw ends on the cut edge. HE / HE / B-01 / B-18.

4 BOX-X 34 × 7 at +4..+11 mm through all 4 layers. Box top stays BELOW +16 (the cheeks).

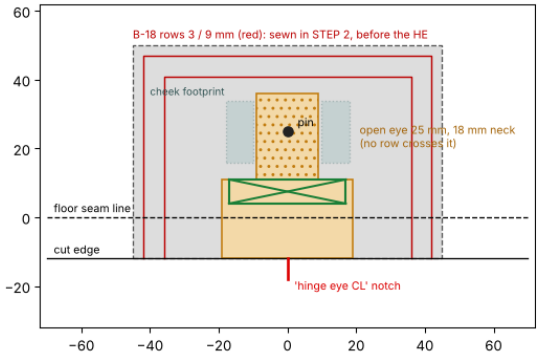
5 BASTE the stack at 6 mm in the allowance. Op 4.15 catches it in the floor seam (zipper / edge foot).

No stitching may cross the eye after step 3.

(c) Loaded on the BL-042 hinge fairlead (pin +25.1; HE-FIT band 22-29)

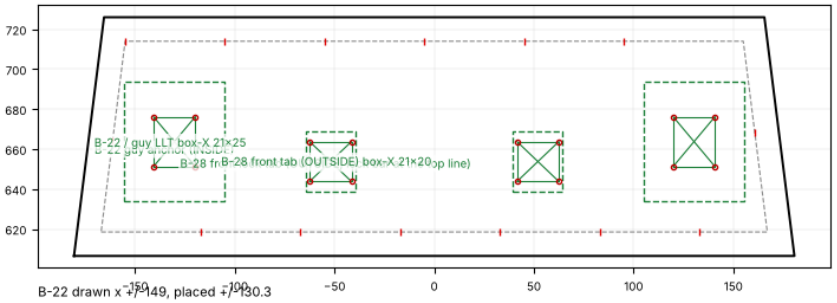


(d) On B-01 from OUTSIDE, x +/-115 (both sides)



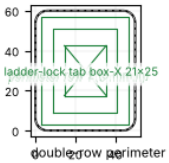
SK-4 Hinge band B-02h: B-22 INSIDE at x +/-130.3 (volatile), LLT box-X, B-28 front tab box-X (outside); then the door hinge seam

B-02h D2 hinge band (z 650 to the collar seam) (cordura_330d_nylon66, cut 1)

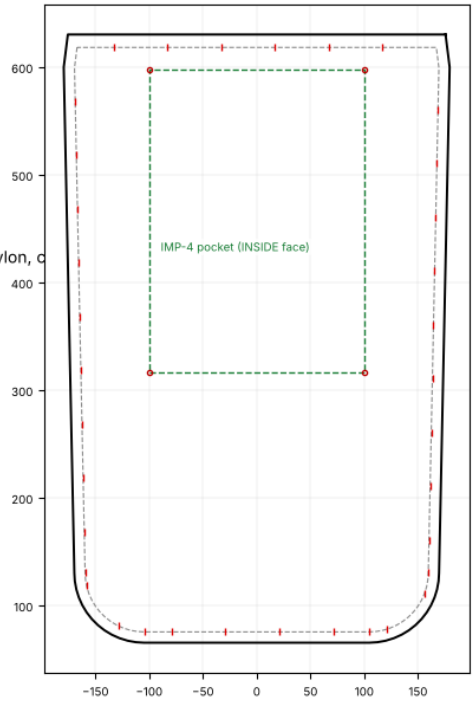


B-22 drawn x +/-149, placed +/-130.3

B-22 guy corner tab (hypalon_coated_nylon, c



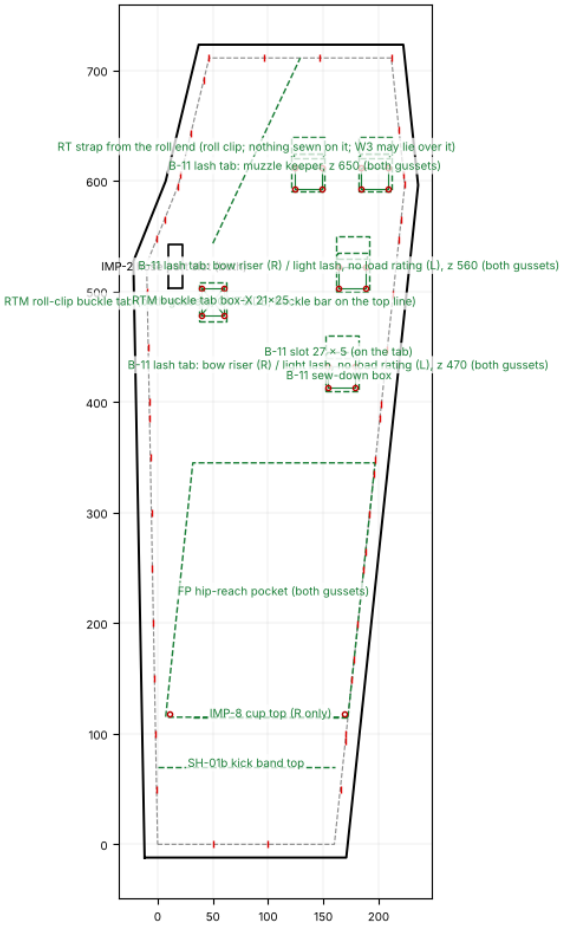
B-02door D2 door (whole aft face z 110-650, R 55 corners) (cordura_330d_nylon



IMP-4 placement INSIDE

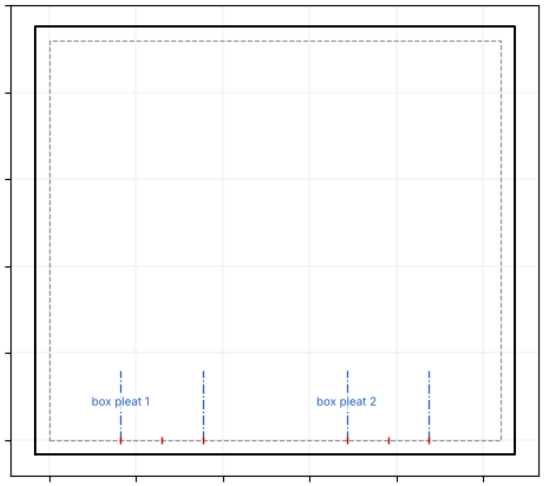
SK-5 Side gusset B-03 (cut L+R): kick band, FP pocket, IMP-8 (R only), hose slot, 4 B-11 lash tabs (op 4.07), K-W keeper notches on the aft edge; corner seams carry HPL (L) and E5 (R)

B-03 side gusset (cordura_330d_nylon66, cut 2)



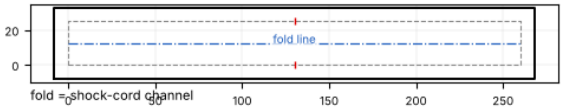
FP z 150-380

FP-a hip-reach pocket body (power mesh) (power_mesh_nylon_spandex, cut 2)



flat 260 wide, 2 box pleats

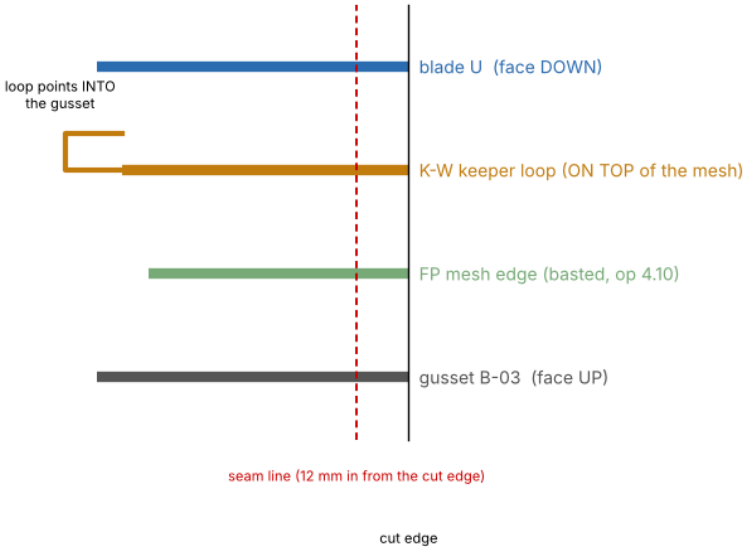
FP-b hip-reach pocket mouth band (camo) (polyester_ht_630d_printab



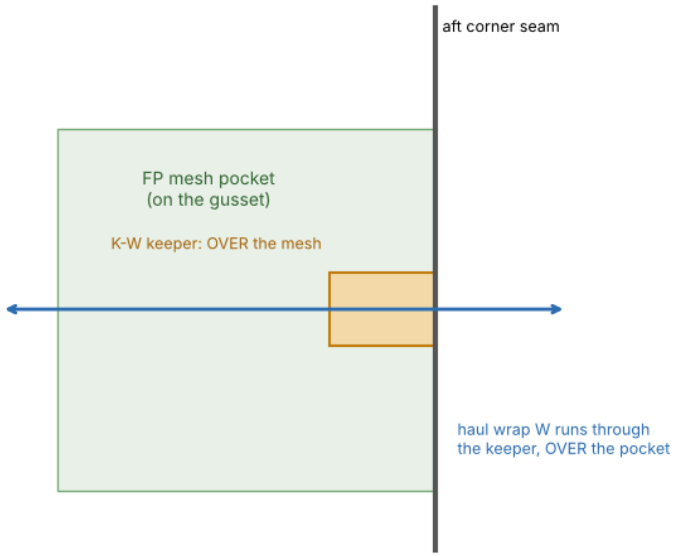
fold = shock-cord channel

SK-9 Aft corner seam (op 4.13): K-W keeper loop goes ON TOP of the FP mesh edge

(a) Before sewing op 4.13: layers bottom to top



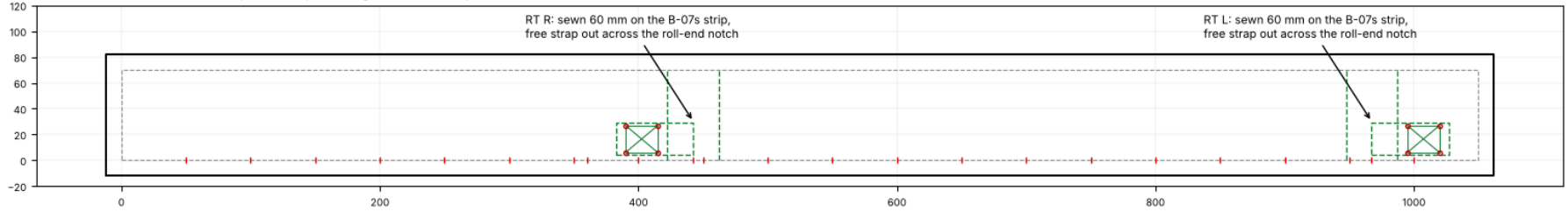
(b) Turned right side out: what you must see



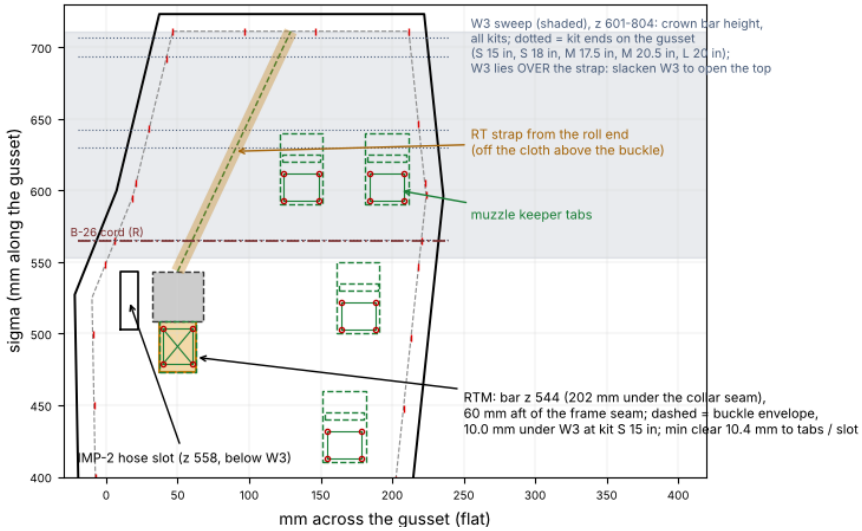
WRONG: keeper under the mesh = the wrap must thread INSIDE the pocket; fixing it means unpicking the 715 mm seam.
Check every keeper before sewing on. Keepers on the FP edge: L z 202 (W1); R z 300 (W2); L z 370 (W2)

SK-10 Roll-top clip: what to sew where (BL-040 B-2; BL-041 RT-FWD; BL-043 W3-OVER / RT-ORIGIN)

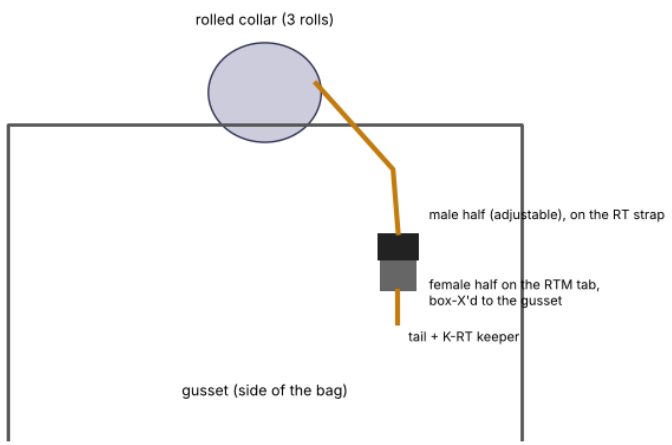
(a) B-07 hem, flat, OUTSIDE up: RT strap rectangles + box-X (op 5.02); red ticks = notches



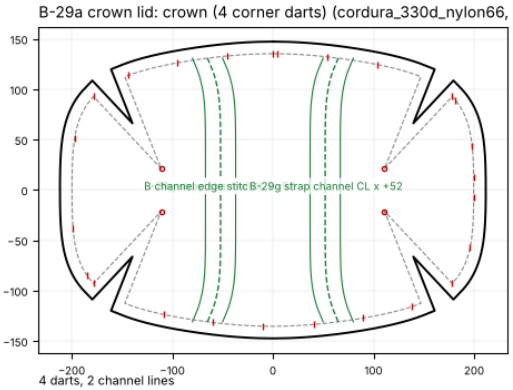
(b) Gusset B-03 (upper part): RTM tab, strap line to the roll end, W3 sweep (op 4.08)

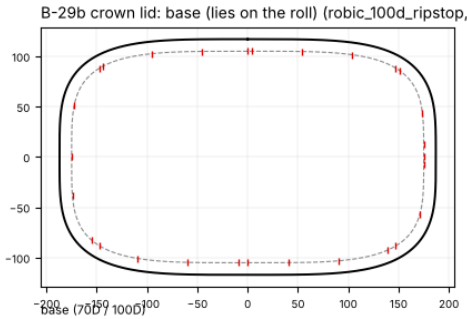


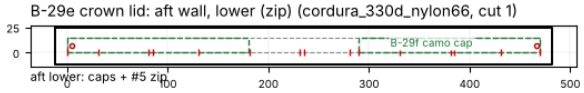
(c) Closed: roll end → RT strap → buckle → RTM tab

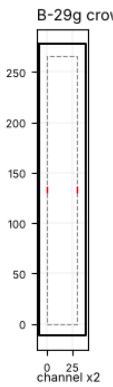


SK-6 B-29 crown lid: darts → channels → zip between the aft strips → caps → crown|wall ring → wall|base ring (turn through the zip)









4. Bill of materials (one bag, full build, crown kit M)

Production qty = 10-bag marker + 3 % lay loss where part A nested one, else net / utilisation (ASSUMPTION). Prototype (1-bag marker) quantity in the Proto column. Waste % = gross / net - 1. Thread: purchase quantity includes the cone waste; the grams in the bag (notes) do not.

Section	Item	Material id	Qty	Unit	Imperial	Waste %	$ / unit	$	Alternate / kit	Proto qty	Source
fabric	challenge_ultra_200: B-01x1 B-06fx1 B-13xx6 B-27fx2	challenge_ultra_200	0.487	m	0.53 yd	14.2	65.6	31.94	same / base	0.743	record challenge_ultra_200; net 0.606 m2 from the pattern areas; 10-bag marker eff 0.92 + 3 % lay loss (part A WASTE_PROD)
fabric	cordura_330d_nylon66: B-03x2 B-02doorx1 B-02hx1 B-07x1 IMP8-ax1 F-02x1 B-29ax1 B-29cx1 B-29dx1 B-29ex1 B-29gx2 A1-ax1	cordura_330d_nylon66	0.824	m	0.90 yd	23.7	15.3	12.61	cordura_330d_nylon66 / ultragrid_210d_uhmwpe_ripstop / addon,as_hunted,base	1.135	record cordura_330d_nylon66; net 1.015 m2 from the pattern areas; 10-bag marker eff 0.88 + 3 % lay loss (part A WASTE_PROD)
fabric	cordura_500d_nylon66: B-05x1 B-16x1 B-16bx1	cordura_500d_nylon66	0.095	m	0.10 yd	32.4	16.4	1.56	challenge_ultra_200 / base	0.36	record cordura_500d_nylon66; net 0.109 m2 from the pattern areas; 10-bag marker eff 0.82 + 3 % lay loss (part A WASTE_PROD)
fabric	hdpe_sheet: B-07sx2 MS-sx2	hdpe_sheet	0.047	m2	0.51 ft2	42.9	4.58	0.21	same / base,meat_trip	0.052	record hdpe_sheet; net 0.033 m2 from the pattern areas; ASSUMPTION UTIL-SMALL 0.70 (small / sheet pieces nested across bags)
fabric	hypalon_coated_nylon: B-17x2 B-18x2 B-22x2 B-11x8	hypalon_coated_nylon	0.059	m2	0.64 ft2	42.9	27.9	1.65	same / base	0.096	record hypalon_coated_nylon; net 0.041 m2 from the pattern areas; ASSUMPTION UTIL-SMALL 0.70 (small / sheet pieces nested across bags)
fabric	polyester_ht_630d_printable: B-02ux1 SH-07x1 SH-01bx2 FP-bx2 B-29fx2 A1-dx1	polyester_ht_630d_printable	0.305	m	0.33 yd	127.6	19.7	6.00	same / addon,as_hunted,base	0.738	record polyester_ht_630d_printable; net 0.201 m2 from the pattern areas; 10-bag marker eff 0.47 + 3 % lay loss (part A WASTE_PROD)
fabric	power_mesh_nylon_spandex: FP-ax2	power_mesh_nylon_spandex	0.129	m	0.14 yd	42.9	10.9	1.41	same / base	0.246	record power_mesh_nylon_spandex; net 0.136 m2 from the pattern areas; ASSUMPTION UTIL-SMALL 0.70 (small / sheet pieces nested across bags)
fabric	robic_100d_ripstop: B-06x1 B-09cx1 IMP4-ax1 IMP4-bx1 B-29bx1 A1-bx1 MS-bx1 MS-cx1	robic_100d_ripstop	0.958	m	1.05 yd	17.8	17.5	16.77	nylon_ripstop_70d / robic_100d_ripstop / addon,as_hunted,base,meat_trip	1.208	record robic_100d_ripstop; net 1.239 m2 from the pattern areas; 10-bag marker eff 0.90 + 3 % lay loss (part A WASTE_PROD)
fabric	silicone_coated_poly_ripstop_40d: F-01ax1 F-01bx2 MS-ax1	silicone_coated_poly_ripstop_40d	1.404	m	1.54 yd	42.9	10.9	15.36	same / as_hunted,meat_trip	1.715	record silicone_coated_poly_ripstop_40d; net 1.498 m2 from the pattern areas; ASSUMPTION UTIL-SMALL 0.70 (small / sheet pieces nested across bags)
fabric	tpu_coated_nylon_840d: B-12x6 B-09px2 IMP8-bx1	tpu_coated_nylon_840d	0.036	m2	0.39 ft2	42.9	16	0.58	same / base	0.09	record tpu_coated_nylon_840d; net 0.025 m2 from the pattern areas; ASSUMPTION UTIL-SMALL 0.70 (small / sheet pieces nested across bags)
fabric_alternate	challenge_ultra_200: B-01x1 B-05x1 B-06fx1 B-13xx6 B-27fx2 B-16x1 B-16bx1	challenge_ultra_200	0.592	m	0.65 yd	17.6	65.6	38.82	challenge_ultra_200 / cordura_500d_nylon66 / base	0.743	record challenge_ultra_200; net 0.715 m2 from the pattern areas; ASSUMPTION UTIL-ALT 0.85
fabric_alternate	cordura_330d_nylon66: IMP8-ax1 F-02x1	cordura_330d_nylon66	0.056	m	0.06 yd	42.9	15.3	0.85	same / base	0.176	record cordura_330d_nylon66; net 0.059 m2 from the pattern areas; ASSUMPTION UTIL-SMALL 0.70 (small / sheet pieces nested across bags)
fabric_alternate	hdpe_sheet: B-07sx2 MS-sx2	hdpe_sheet	0.047	m2	0.51 ft2	42.9	4.58	0.21	same / base,meat_trip	0.052	record hdpe_sheet; net 0.033 m2 from the pattern areas; ASSUMPTION UTIL-SMALL 0.70 (small / sheet pieces nested across bags)
fabric_alternate	hypalon_coated_nylon: B-17x2 B-18x2 B-22x2 B-11x8	hypalon_coated_nylon	0.059	m2	0.64 ft2	42.9	27.9	1.65	same / base	0.096	record hypalon_coated_nylon; net 0.041 m2 from the pattern areas; ASSUMPTION UTIL-SMALL 0.70 (small / sheet pieces nested across bags)

Section	Item	Material id	Qty	Unit	Imperial	Waste %	$ / unit	$	Alternate / kit	Proto qty	Source
fabric_alternate	nylon_ripstop_70d: B-06x1 B-09cx1 B-29bx1 A1-bx1 MS-bx1 MS-cx1	nylon_ripstop_70d	0.892	m	0.98 yd	17.6	11.8	10.48	robic_100d_ripstop / addon,as_hunted,base,meat_trip	1.129	record nylon_ripstop_70d; net 1.117 m2 from the pattern areas; ASSUMPTION UTIL-ALT 0.85
fabric_alternate	polyester_ht_630d_printable: B-02ux1 SH-07x1 SH-01bx2 FP-bx2 B-29fx2 A1-dx1	polyester_ht_630d_printable	0.158	m	0.17 yd	17.6	19.7	3.10	same / addon,as_hunted,base	0.738	record polyester_ht_630d_printable; net 0.201 m2 from the pattern areas; ASSUMPTION UTIL-ALT 0.85
fabric_alternate	power_mesh_nylon_spandex: FP-ax2	power_mesh_nylon_spandex	0.129	m	0.14 yd	42.9	10.9	1.41	same / base	0.246	record power_mesh_nylon_spandex; net 0.136 m2 from the pattern areas; ASSUMPTION UTIL-SMALL 0.70 (small / sheet pieces nested across bags)
fabric_alternate	robic_100d_ripstop: IMP4-ax1 IMP4-bx1	robic_100d_ripstop	0.115	m	0.13 yd	42.9	17.5	2.02	same / base	0.296	record robic_100d_ripstop; net 0.123 m2 from the pattern areas; ASSUMPTION UTIL-SMALL 0.70 (small / sheet pieces nested across bags)
fabric_alternate	silicone_coated_poly_ripstop_40d: F-01ax1 F-01bx2 MS-ax1	silicone_coated_poly_ripstop_40d	1.157	m	1.27 yd	17.6	10.9	12.65	same / as_hunted,meat_trip	1.715	record silicone_coated_poly_ripstop_40d; net 1.498 m2 from the pattern areas; ASSUMPTION UTIL-ALT 0.85
fabric_alternate	tpu_coated_nylon_840d: B-12x6 B-09px2 IMP8-bx1	tpu_coated_nylon_840d	0.036	m2	0.39 ft2	42.9	16	0.58	same / base	0.09	record tpu_coated_nylon_840d; net 0.025 m2 from the pattern areas; ASSUMPTION UTIL-SMALL 0.70 (small / sheet pieces nested across bags)
fabric_alternate	ultragrid_210d_uhmwpe_ripstop: B-03x2 B-02doorx1 B-02hx1 B-07x1 B-29ax1 B-29cx1 B-29dx1 B-29ex1 B-29gx2 A1-ax1	ultragrid_210d_uhmwpe_ripstop	0.75	m	0.82 yd	17.6	21.9	16.40	cordura_330d_nylon66 / addon,as_hunted,base	0.963	record ultragrid_210d_uhmwpe_ripstop; net 0.956 m2 from the pattern areas; ASSUMPTION UTIL-ALT 0.85
fabric_print	camo print, Owner Base accent 380, placed (3 sets/yd) -- see camo/README.md (print file camo/print/, colours camo/colour_spec.csv, placement camo/placement.csv, vendors camo/vendors.csv)	polyester_300d_printable	0.4	yd		0	30.2	12.10	polyester_ht_630d_printable (unprinted, fabric line) / base	0	camo/bom_print_line_proposed.csv: RBTR HyperD 300 DYO $30.25/yd, 1 yd min (vendors.csv)
webbing	elastic_25mm: K-28 K-G K-IM K-RT	elastic_25mm	1.04	m	1.14 yd	5	1.1	1.14	same / as_hunted,base		record elastic_25mm
webbing	elastic_6mm: F-EL	elastic_6mm	2.558	m	2.80 yd	5	0.6	1.53	same / as_hunted		record elastic_6mm
webbing	poly_webbing_25mm: B-28 B-28s B-28t BRL BRS E5 HPL I2K I2T IM1 IM2 IM3 IMC K-W LLT RT RTM TG US US2 USA W1 W2 W3 WX	poly_webbing_25mm	17.36	m	18.99 yd	5	0.7	12.15	same / as_hunted,base,crown kit M,crown kit M (as hunted),meat_trip		record poly_webbing_25mm
webbing	poly_webbing_38mm: B-24 B-25 HE IME T3	poly_webbing_38mm	3.25	m	3.55 yd	5	1	3.25	same / base,meat_trip		record poly_webbing_38mm
webbing	shock_cord_3mm: B-26 E5-cord FP-cord MK-cord	shock_cord_3mm	1.826	m	2.00 yd	5	0.5	0.91	same / base		record shock_cord_3mm
hardware	#10 coil zip chain, door D2 (2 sliders)	coil_zipper_10	1.418	m	1.55 yd	0	4	5.67	same / base		record coil_zipper_10; ledger rows: D2 door zip
hardware	#10 coil slider, metal pull (door)	slider_10	2	unit		0	1.99	3.98	same / base		record slider_10; ledger rows: D2 door zip
hardware	#5 coil zip chain: lid, IMP-4, F-02 (3 zips)	coil_zipper_5	0.97	m	1.06 yd	0	2	1.94	same / base,as_hunted		record coil_zipper_5; ledger rows: -
hardware	#5 coil slider + pull (3: lid, IMP-4, F-02)	slider_5	3	unit		0	0.96	2.88	same / base,as_hunted		record slider_5; ledger rows: BL-034 records: F-02 fly pocket #5 slider
hardware	door pull cord 3 mm Dyneema, 2 x 0.15 m (ASSUMPTION length)	dyneema_cord_3mm	0.3	m	0.33 yd	0	0.9	0.27	same / base		record dyneema_cord_3mm; ledger rows: door: 2 cord pulls
hardware	rubber pull sleeve	rubber_pull_sleeve	2	unit		0	0.15	0.30	same / base		ASSUMPTION; ledger rows: door: 2 cord pulls
hardware	rated 25 mm aluminium G-hook: IM1, IM2 (2) + W1-W3 (3) + bow riser (1)	g_hook_25mm_aluminium	6	unit		0	4	24.00	same / base		record g_hook_25mm_aluminium; ledger rows: BL-010 S9; IM-hooks; W-hw
hardware	rated 25 mm aluminium G-hook: W haul extenders	g_hook_25mm_aluminium	3	unit		0	4	12.00	same / meat_trip		record g_hook_25mm_aluminium; ledger rows: -

Section	Item	Material id	Qty	Unit	Imperial	Waste %	$ / unit	$	Alternate / kit	Proto qty	Source
hardware	wiregate carabiner 22 kN: IM3 bag hooks (2) + IM3 meat free end (1)	carabiner_wiregate_22kn	3	unit		0	10	30.00	same / base		record carabiner_wiregate_22kn; ledger rows: BL-010 A1; BL-010 S2; attachment system: bag hooks 1500; attachment: bag hooks, 38 mm
hardware	wiregate carabiner 22 kN: B-24 float links	carabiner_wiregate_22kn	2	unit		0	10	20.00	same / meat_trip		record carabiner_wiregate_22kn; ledger rows: B-24-hooks; BL-010 S12
hardware	heavy 25 mm ladder lock (9): B-28 (4) + guys (2) + W1-W3 (3)	ladder_lock_25mm_heavy	9	unit		0	2.1	18.90	same / base,as_hunted		record ladder_lock_25mm_heavy; ledger rows: BL-010 A3; BL-010 S9; W-hw
hardware	25 mm aluminium cam buckle: IM1-IM3 (3) + US1/US2 (2)	cam_25mm_al	5	unit		0	25.5	127.50	same / base		record cam_25mm_al; ledger rows: IM-cams; US-cams
hardware	25 mm STAINLESS tri-glide: US1/US2 cradle backstop behind the cam (2; orchestrator 2026-11-13, 11 g, 3.75 kN)	triglide_25mm	2	unit		0	2.84	5.68	same / base		record triglide_25mm stainless_alt (EUR 3.10 incl. VAT, converted as the record); ledger rows: BL-034 records: US1/US2 backstop tri-glides; US-tg
hardware	25 mm nylon tri-glide: guy tail keepers (2)	triglide_25mm	2	unit		0	0.37	0.74	same / base		record triglide_25mm; ledger rows: BL-034 records: guy tri-glides
hardware	POM toggle 30 x 8 x 5 (4): B-28 rear (2) + guy loops (2)	pom_toggle	4	unit		0	0.12	0.48	same / base,as_hunted		record pom_toggle; ledger rows: BL-004 REC: guy toggles
hardware	25 mm side-release buckle (roll top)	buckle_side_release_25mm	2	unit		0	0.8	1.60	same / base		record buckle_side_release_25mm; ledger rows: -
hardware	spring cord lock (3): cord lock (E-5) + rifle muzzle keepers (2, BL-035)	cord_lock_spring	3	unit		0	0.4	1.20	same / base		record cord_lock_spring; ledger rows: -
hardware	TPU sleeve on the IM3 carabiner (2)	tpu_hook_sleeve	2	unit		0	0.2	0.40	same / base		ASSUMPTION; ledger rows: attachment: TPU hook sleeves
hardware_frame_side	printed fairleads (12 stations, in the H1 print; the US2 outer cheeks are the bolt-on clevises), 7075 pins (10: IM1-IM3 and US1 / US2, 36 mm shank), Ti-6Al-4V HINGE pins (2: 5 mm h8 ground, 44 mm, clip groove; BL-049, NOT 7075), UHMWPE sleeves (12), lifter keepers; crown pawl = the LOW-PROFILE pawl print/frame_h1/crown_pawl_lp_fdm_pacf (P1) / _lp_slm (production), NOT the old crown_pawl_* (it does not fit H1, BL-047)	frame kit	1	set		0			/ base		ledger rows: BL-034 records: lifter tri-glide keepers; BL-044: hinge pins; US-anchor; attachment: 7075 pins; attachment: UHMWPE sleeves; attachment: lifter tri-glide; attachment: printed fairleads
hardware_frame_side	z 234 JOINT: Ti gr5 low-head socket cap screw M5 x 45 DIN 7984 (NOT the M5 x 40 of the US2 clevis: too short here)	-	4	unit		0			/ frame kit		ledger rows: -
hardware_frame_side	z 234 JOINT: M5 hex nut A2 ISO 4032 (8 AF x 4)	-	4	unit		0			/ frame kit		ledger rows: -
hardware_frame_side	z 234 JOINT: M5 washer 12 mm OD x 1.0 A2 (12 mm is LOAD-BEARING; NOT the 10 mm clevis washer)	-	8	unit		0			/ frame kit		ledger rows: -
hardware_frame_side	US2 CLEVIS (BL-050): Ti gr5 low-head socket cap screw M5 x 40 DIN 7984, 2 per side (NOT the M5 x 45 of z 234: it would stand 5 mm longer into the trap)	-	4	unit		0			/ frame kit		ledger rows: -
hardware_frame_side	US2 CLEVIS (BL-050): M5 hex nut A2 ISO 4032 (8 AF x 4), side traps	-	4	unit		0			/ frame kit		ledger rows: -
hardware_frame_side	US2 CLEVIS (BL-050): M5 washer 10 mm OD x 1.0 A2 DIN 433 (12 mm does not fit the clevis block; NOT for z 234)	-	8	unit		0			/ frame kit		ledger rows: -

Section	Item	Material id	Qty	Unit	Imperial	Waste %	$ / unit	$	Alternate / kit	Proto qty	Source
thread	bonded_nylon_tex135 (needle + bobbin)	bonded_nylon_tex135	35.1	m	38.39 yd	15	0.0081	0.28	same / all		record bonded_nylon_tex135; sum of op stitch lengths x 2.75 (ASSUMPTION THREAD-FACTOR) + 15% cone waste on the purchase quantity only
thread	bonded_nylon_tex70 (needle + bobbin)	bonded_nylon_tex70	82	m	89.68 yd	15	0.0052	0.43	same / all		record bonded_nylon_tex70; sum of op stitch lengths x 2.75 (ASSUMPTION THREAD-FACTOR) + 15% cone waste on the purchase quantity only
thread	bonded_nylon_tex90 (needle + bobbin)	bonded_nylon_tex90	124.4	m	136.05 yd	15	0.0061	0.76	same / all		record bonded_nylon_tex90; sum of op stitch lengths x 2.75 (ASSUMPTION THREAD-FACTOR) + 15% cone waste on the purchase quantity only
consumable	6 mm double-sided basting tape (zips)	basting_tape_6mm	3.66	m	4.00 yd	0	0.05	0.18	/		ASSUMPTION
consumable	labels: size / care / ID, hang tag, load-limit warning card	label_set	1	set		0	0.6	0.60	/		ASSUMPTION: size / care / ID label + hang tag + warning card
consumable	silicone seam sealer for the silpoly fly and liner seams, ~10 ml (ASSUMPTION)	seam_sealer	1	bag		0	0.8	0.80	/		ASSUMPTION: 10 ml of a silicone seam sealer
consumable	urethane seam sealer for the B-16b channel tail stitch lines on the PU-coated floor B-05, ~3 ml (BL-050 B; ASSUMPTION)	seam_sealer_pu	1	bag		0	0.4	0.40	/		ASSUMPTION: about 3 ml of a urethane seam sealer (2 x 70 mm x 2 lines + 4 bartacks)
packaging	poly bags + carton share	polybag_carton	1	bag		0	0.9	0.90	/		ASSUMPTION: 2 recycled poly bags + 1/10 carton

5. Cutting plan (pattern list, markers, cut lengths)

Markers: ../patterns/markers/. Cut windows, slots and slits first; knife clockwise from the grain arrow; notches 6 mm. Ply height: woven nylon 24, laminates / mesh / silpoly 12, camo single ply (one way, print placement), sheet goods laser single sheet.

Set	Material	Pieces	1-bag m	eff	10-bag m / bag	eff	Lay	Prototype	Production
primary	challenge_ultra_200	B-01x1 B-06fx1 B-13xx6 B-27fx2	0.743	0.5838	0.4726	0.9177	laminate face up, all pieces one way	rotary; windows and B-27 slits with a hot knife / scalpel on a cutting mat	straight knife, 12 plies max (laminate creeps), or laser single ply
primary	cordura_330d_nylon66	B-03x2 B-02doorx1 B-02hx1 B-07x1 IMP8-ax1 F-02x1 B-29ax1 B-29cx1 B-29dx1 B-29ex1 B-29gx2 A1-ax1	1.1348	0.6178	0.7996	0.8767	face up, one way not needed	rotary + hot-knife the window / slot cut-outs	straight knife, 10-bag marker, 24 plies max; or laser single ply (sealed edges)
primary	cordura_500d_nylon66	B-05x1 B-16x1 B-16bx1	0.36	0.2094	0.0921	0.8186	face up	rotary	straight knife, 24 plies max
primary	hdpe_sheet	B-07sx2 MS-sx2	0.043	0.6213			free rotation	shear / router, round every corner R >= 3	CNC router from the DXF
primary	hypalon_coated_nylon	B-17x2 B-18x2 B-22x2 B-11x8	0.07	0.4387			coated face up, free rotation	laser from the DXF (makerspace) or a knife + template	laser single sheet; dies above 500 bags
primary	polyester_ht_630d_printable	B-02ux1 SH-07x1 SH-01bx2 FP-bx2 B-29fx2 A1-dx1	0.738	0.1878	0.2961	0.468	print face up, ONE WAY	rotary	single ply (printed camo, one way, pattern placement)
primary	power_mesh_nylon_spandex	FP-ax2	0.246	0.3807			stretch ACROSS, one way	rotary, no stretch while cutting	straight knife 12 plies, relaxed 24 h before cutting
primary	robic_100d_ripstop	B-06x1 B-09cx1 IMP4-ax1 IMP4-bx1 B-29bx1 A1-bx1 MS-bx1 MS-cx1	1.208	0.6966	0.9305	0.9043	face up	rotary	straight knife, 24 plies max (paper between every 6 plies: slippery)
primary	silicone_coated_poly_ripstop_40d	F-01ax1 F-01bx2 MS-ax1	1.7146	0.5932			face up	rotary, weights not pins	straight knife 12 plies with paper underlay
primary	tpu_coated_nylon_840d	B-12x6 B-09px2 IMP8-bx1	0.06	0.2887			coated face up, free rotation	laser from the DXF	laser single sheet
alternate	challenge_ultra_200	B-01x1 B-05x1 B-06fx1 B-13xx6 B-27fx2 B-16x1 B-16bx1	0.743	0.6889			laminate face up, all pieces one way	rotary; windows and B-27 slits with a hot knife / scalpel on a cutting mat	straight knife, 12 plies max (laminate creeps), or laser single ply
alternate	cordura_330d_nylon66	IMP8-ax1 F-02x1	0.176	0.2327			face up, one way not needed	rotary + hot-knife the window / slot cut-outs	straight knife, 10-bag marker, 24 plies max; or laser single ply (sealed edges)
alternate	hdpe_sheet	B-07sx2 MS-sx2	0.043	0.6213			free rotation	shear / router, round every corner R >= 3	CNC router from the DXF
alternate	hypalon_coated_nylon	B-17x2 B-18x2 B-22x2 B-11x8	0.07	0.4387			coated face up, free rotation	laser from the DXF (makerspace) or a knife + template	laser single sheet; dies above 500 bags
alternate	nylon_ripstop_70d	B-06x1 B-09cx1 B-29bx1 A1-bx1 MS-bx1 MS-cx1	1.129	0.6714			face up	rotary	straight knife, 12 plies max (slippery)
alternate	polyester_ht_630d_printable	B-02ux1 SH-07x1 SH-01bx2 FP-bx2 B-29fx2 A1-dx1	0.738	0.1878			print face up, ONE WAY	rotary	single ply (printed camo, one way, pattern placement)
alternate	power_mesh_nylon_spandex	FP-ax2	0.246	0.3807			stretch ACROSS, one way	rotary, no stretch while cutting	straight knife 12 plies, relaxed 24 h before cutting
alternate	robic_100d_ripstop	IMP4-ax1 IMP4-bx1	0.296	0.2819			face up	rotary	straight knife, 24 plies max (paper between every 6 plies: slippery)
alternate	silicone_coated_poly_ripstop_40d	F-01ax1 F-01bx2 MS-ax1	1.7146	0.5932			face up	rotary, weights not pins	straight knife 12 plies with paper underlay
alternate	tpu_coated_nylon_840d	B-12x6 B-09px2 IMP8-bx1	0.06	0.2887			coated face up, free rotation	laser from the DXF	laser single sheet
alternate	ultragrid_210d_uhmwpe_ripstop	B-03x2 B-02doorx1 B-02hx1 B-07x1 B-29ax1 B-29cx1 B-29dx1 B-29ex1 B-29gx2 A1-ax1	0.963	0.6729			face up	rotary (UHMWPE grid: new blade)	straight knife 12 plies; UHMWPE dulls blades: sharpen every lay

Webbing, cord and elastic cut lengths

ID	Use	Material	Length mm (in)	Qty	Kit	Source
B-28	B-28 crown strap, kit S	poly_webbing_25mm	1068 (42.1)	2	crown kit S (as hunted)	corrie_f.crown_kit_straps
B-28	B-28 crown strap, kit M	poly_webbing_25mm	1007 (39.6)	2	crown kit M (as hunted)	corrie_f.crown_kit_straps
B-28	B-28 crown strap, kit L	poly_webbing_25mm	948 (37.3)	2	crown kit L (as hunted)	corrie_f.crown_kit_straps
B-28t	lock tab	poly_webbing_25mm	60 (2.4)	4	as_hunted	corrie_f.CROWN_TAB, 2 per strap (front + rear lock)
B-28s	toggle loop	poly_webbing_25mm	60 (2.4)	2	as_hunted	corrie_f.CROWN_TAB, 1 per strap (the B-25 second fold)
K-28	B-28 tail keeper (25 mm elastic loop)	elastic_25mm	90 (3.5)	4	as_hunted	corrie_f.crown_kit_straps keepers_per_strap x 2
W1	haul wrap W1 (cut for carry)	poly_webbing_25mm	750 (29.5)	1	base	strap_tails.wrap_rows
W2	haul wrap W2 (cut for carry)	poly_webbing_25mm	822 (32.4)	1	base	strap_tails.wrap_rows
W3	haul wrap W3, kit S	poly_webbing_25mm	1034 (40.7)	1	crown kit S	strap_tails.wrap_rows
W3	haul wrap W3, kit M	poly_webbing_25mm	987 (38.8)	1	crown kit M	strap_tails.wrap_rows
W3	haul wrap W3, kit L	poly_webbing_25mm	935 (36.8)	1	crown kit L	strap_tails.wrap_rows
WX	W haul extender (meat-trip kit)	poly_webbing_25mm	740 (29.1)	3	meat_trip	strap_tails.wrap_rows haul extender_mm
K-W	W gore keeper loop (4, in the aft corner seams: W1, W2)	poly_webbing_25mm	80 (3.1)	4	base	ledger 'keepers: 6 haul-wrap keeper loops' (ASSUMPTION 80 mm cut) less the 2 W3 loops (BL-044 DECISION KW-W3-OFF row)
RT	roll-top side clip strap	poly_webbing_25mm	565 (22.2)	2	base	strap_tails.roll_row
RTM	roll-clip buckle tab on the gusset (female half)	poly_webbing_25mm	70 (2.8)	2	base	BL-040 DECISION RT-MATE (LLT tab rule); BL-041 RT-FWD placement (rt_layout)
K-RT	roll clip tail keeper (25 mm elastic)	elastic_25mm	90 (3.5)	2	base	BL-031
IM1	internal meat strap IM1	poly_webbing_25mm	990 (39.0)	1	base	strap_tails.im_rows (unchanged)
IM2	internal meat strap IM2	poly_webbing_25mm	1023 (40.3)	1	base	strap_tails.im_rows (unchanged)
IM3	internal meat strap IM3	poly_webbing_25mm	1083 (42.6)	1	base	strap_tails.im_rows (unchanged)
IMC	IM cam anchor tab (cam bar <-> hook / carabiner)	poly_webbing_25mm	192 (7.6)	3	base	BL-041 DECISION IM-CAM-TAB (review m-19 / n-15: the cam had no sewn anchor)
K-IM	IM tail keeper (25 mm elastic)	elastic_25mm	90 (3.5)	3	base	BL-031
IME	IM fairlead eye (38 mm)	poly_webbing_38mm	200 (7.9)	6	base	ledger IM-eyes 6 x 0.20 m
US	under-strap US1 (pins z 95)	poly_webbing_25mm	550 (21.7)	1	base	corrie_f.US_CUT_MM (BL-050 B; us_route.json strap_lengths)
US2	under-strap US2 (pins z 142)	poly_webbing_25mm	925 (36.4)	1	base	corrie_f.US_CUT_MM (BL-050 B; us_route.json strap_lengths)
USA	US cam anchor (right pin <-> cam bar)	poly_webbing_25mm	202 (8.0)	2	base	BL-044 (BL-045: the US cam had no sewn anchor); IMC construction
B-24	float stand-off link (38 mm, meat-trip)	poly_webbing_38mm	435 (17.1)	2	meat_trip	attach_system.link_piece; ledger B-24-web 2 x 435 mm
HE	hinge eye (38 mm)	poly_webbing_38mm	96 (3.8)	2	base	BL-040 DECISION HE-ONE + BL-041 HE-PIN + BL-044 DECISION HE-ENV (cut 38 x 96; eye solved by he_geometry on bag_attach.HINGE_EYE_ENVELOPE)
T3	IM3 bag-hook tongue (38 mm)	poly_webbing_38mm	160 (6.3)	2	base	attach_system.drawing_pieces T2/T3 (IM2 hooks removed by BL-004: 2 left)

ID	Use	Material	Length mm (in)	Qty	Kit	Source
B-25	guy bridle on the crown top bar (38 mm, BL-041 / BL-043 topology)	poly_webbing_38mm	513 (20.2)	1	base	corrie_f.b25_topology (BL-043 B25-TOPO) on corrie_f.guy_bridle(wide=True) (crown_tube_guy_loop.md 5, 8.1)
TG	port guy (B-25 toggle -> B-27 port -> B-22 inside)	poly_webbing_25mm	500 (19.7)	2	base	attach_system.GUY cut_mm
LLT	guy ladder-lock tab on B-22	poly_webbing_25mm	70 (2.8)	2	base	attach_system.drawing_pieces LLT
K-G	guy tail keeper (25 mm elastic)	elastic_25mm	90 (3.5)	2	base	ledger 'guy tail elastic keepers x2'
HPL	scope-pod hard-point loop	poly_webbing_25mm	80 (3.1)	4	base	corrie_f.LOOP_M
E5	pole grip tab (cord lock)	poly_webbing_25mm	60 (2.4)	1	base	evolve.POLE tab_m
I2T	IMP-2 hang tab (B-06f)	poly_webbing_25mm	100 (3.9)	1	base	lean.fell IMP-2L 0.10 m
I2K	IMP-2 bottom keeper tab (B-01)	poly_webbing_25mm	60 (2.4)	2	base	lean.fell IMP-2L 0.06 m
BRL	bow riser loop (through B-11)	poly_webbing_25mm	100 (3.9)	2	base	ledger bow riser carry (ASSUMPTION 100 mm loops)
BRS	bow riser strap + G-hook	poly_webbing_25mm	300 (11.8)	1	base	ledger bow riser carry (ASSUMPTION 300 mm strap)
B-26	B-26 tripod keeper, 3 mm shock cord (ends 15 mm in the R corner seams)	shock_cord_3mm	201 (7.9)	1	base	corrie_f.tripod_keeper
FP-cord	FP mouth shock cord	shock_cord_3mm	320 (12.6)	2	base	corrie_f.hip_reach_pockets (w + pleat + 60)
E5-cord	pole quiver grip cord	shock_cord_3mm	300 (11.8)	1	base	evolve.POLE
MK-cord	rifle muzzle keeper, 3 mm shock cord loop through the 2 B-11 muzzle tabs + spring cord lock (one per gusset)	shock_cord_3mm	299 (11.8)	2	base	corrie_f.muzzle_keeper (ASSUMPTION MUZZLE-CORD)
F-EL	fly hem elastic, 6 mm (spine top + bottom, sides bottom + frame)	elastic_6mm	2436 (95.9)	1	as_hunted	corrie_f.fly_pieces (FLY_ELASTIC_RATIO x hem channels)

6. Sewing operation sequence

All flat work before the bag becomes 3D. Clips, never pins, in coated fabric. Swatch every new stack. Load points only after their proof test passes (section 8). V = volatile (re-run on a pattern change).

Seq	Operation	How (beginner)	Pieces	Machine	Needle	Thread	SPI	Stitch / seam	mm	Ref	h / SAM min
0.01	Print and check the test square	Print patterns/print_letter/00_PRINT_FIRST_test_square.pdf at 100 %. The solid square must read 100 mm (+/- 1 mm) both ways, the dashed one 4 in. Do not cut from a wrong-scale print.		bench	-	-	-	- / -		QC-01	0.10 / 0.0 1x
0.02	Machine set-up and stack swatches	Make a 150 x 150 mm swatch of each stack and set tension until the lock sits mid-stack on both faces: (a) ULTRA + 330D body seam, tex 90 / Nm 110 R, 8 SPI; (b) Hypalon on ULTRA, double row; (c) 25 mm box-X tex 135 / Nm 140 R, 7 SPI; (d) 70D / mesh at 10 SPI. Keep them in the QC folder.		WF1	Nm 110/18 DPx17, round point R	BN tex90	8	301 / SSa2, LSd, BOX, SSa (swatches)		SK-1, SK-2; QC-05, QC-06	1.00 / 0.0 1x
0.03	Sew the first-article load samples	Before cutting the bag, sew the proof specimens exactly as the bag: 10 x 25 mm box-X loops + 2 IMC cam tabs (MT-1), 5 T3 tongues on 70 mm B-17 tabs on B-01 swatches (PT-2), 1 hinge eye for the HE-FIT dry fit on a printed z 60 fairlead FIRST, then 5 hinge eyes exactly as op 4.05 and SK-8 in its order (B-18 rows first through B-18 + B-01 only; then HE folded in half, eye UP, box-X 34 x 7 through HE + B-01 + B-18 below the cheek line; B-18 and both HE raw ends caught in a B-01 | B-05 SSa2 floor seam sewn with the zipper / edge foot, no binding: PT-3), 5 B-24 links (PT-6), 5 TL-1 sleeves (38 mm B-25) on real crown bar stock. The bag's load points are sewn only after these pass.		BT / PS (prototype WF2 + J1)	Nm 140/22 DPx17, round point R	BN tex135	7	301 box-X path / BOX + LSd (as the bag)		SK-1; TEST gates; all load points	3.00 / 0.0 1x
0.04	Cut 77 fabric / sheet pieces	Per cutting_plan.csv: lay face up (one way for camo and mesh), L+R pieces face to face, cut windows, slots and slits FIRST, knife clockwise from the grain arrow. Notches 6 mm, never past the seam line. Hypalon, TPU-840D and HDPE from the DXF on the laser / router.	all	SK / LASER / HK	-	-	-	- / -		QC-02, QC-03, QC-04	9.24 / 66.5
0.05	Hot-knife 78 webbing / elastic / cord pieces	Cut lengths in cutting_plan.csv (crown kit M shown; B-28 and W3 per crown kit). Use the J8 stop; seal both ends; no hard bead wider than the webbing.	webbing	SK / LASER / HK	-	-	-	- / -		QC-04	2.34 / 16.9
0.06	Mark every placement	Chalk / silver pen on the INSIDE unless the pattern says OUTSIDE; awl the drill marks (box corners, slit ends, window corners, zip stops). Pieces: B-01, B-03 (L only / R only marks), B-02h, B-02door, B-02u, B-06f, B-29a.		bench	-	-	-	- / -			0.60 / 4.3
1.01 V	FP hip-reach pocket: pleats	Fold the 2 box pleats on each FP-a (flat 260 mm wide, finished 165 mm), baste across the BOTTOM edge only, at 6 mm. The top stays 260 mm wide (op 1.02 sews the band to it; the cord gathers it). Do not stretch the mesh.	FP-a x2	WF1	Nm 90/14 DPx17, medium ball SUK (knit mesh)	BN tex70	10	301 / SSa	520	SK-5	0.71 / 5.1
1.02 V	FP mouth band and shock-cord channel	Sew FP-b to the FP-a top (260 mm), fold the band on its fold line over the seam, edge-stitch 2 mm to form the channel; thread the shock cord (FP-cord) and bar it at both ends inside the side allowance.	FP-a, FP-b x2, FP-cord	WF1	Nm 90/14 DPx17, medium ball SUK (knit mesh)	BN tex70	10	301 / SSa + EFa channel	1040	SK-2	0.62 / 4.5
1.03	IMP-4 zip pocket	Set the #5 zip (210 mm) in the top of IMP4-a (basting tape, ZIP seam), stops at the drill marks; sew the bellows strip IMP4-b round bottom + sides of IMP4-a.	IMP4-a, IMP4-b, #5 zip	WF1	Nm 90/14 DPx17, round point R	BN tex70	10	301 / ZIP + SSa	1240		2.29 / 16.5
1.04	IMP-8 cam cup	Fold IMP8-a, sew the 2 side seams; sew the laser-cut TPU base IMP8-b in (TPU face out).	IMP8-a, IMP8-b	WF1	Nm 110/18 DPx17, round point R	BN tex90	8	301 / SSa	610		0.75 / 5.4
1.05	IMP-2 hydration sleeve	Fold B-09c into a tube, one side seam; turn the top edge 10 mm and edge-stitch; I2T hang tab goes in later (op 5.01).	B-09c	WF1	Nm 90/14 DPx17, round point R	BN tex70	10	301 / SSa + EFa	930		0.56 / 4.0

Seq	Operation	How (beginner)	Pieces	Machine	Needle	Thread	SPI	Stitch / seam	mm	Ref	h / SAM min
1.06 V	F-02 fly base pocket	Set the #5 zip (230 mm), turn the edges 10 mm. It is sewn to the floor B-05 in step 4 on the drawn 'F-02 fly pocket' placement (190 x 140 mm, OUTSIDE, zip on the aft edge, corners drilled; pattern ASSUMPTION F02-CENTRE fore-aft).	F-02, #5 zip	WF1	Nm 110/18 DPx17, round point R	BN tex90	8	301 / ZIP + EFa	1120		1.13 / 8.1
2.01	B-12 strap-port patches on the swell crests	Lay a B-12 (TPU face out) on the crest of each B-13x (6, not the hinge swells), centred on the swell's 'port slot' mark, and sew the 3 mm perimeter. Then cut the swell fabric out through the B-12's laser-cut 45 x 12 slot (scalpel, the B-12 slot is the guide; the swell slot is a mark on the pattern, not a pre-cut hole) and edge-stitch the slot.	B-12 x6, B-13x x6	WF2	Nm 110/18 DPx17, round point R	BN tex90	8	301 / LSd	1843	SK-2	1.47 / 10.6
2.02 V	Box the swells	Sew the 4 corner folds of every swell (B-13x x6): fold on 'box the corner', sew across. Practice: set one swell into a scrap window before touching B-01.	B-13x x6	WF1	Nm 110/18 DPx17, round point R	BN tex90	8	301 / SSa			1.50 / 10.8
2.03	T3 IM3 tongues onto B-17	Fold each T3 45 mm round the bed (empty loop: the carabiner clips in later), box-X the loop (34 x 30); box-X the other end (34 x 34) through T3 + B-17 on the drawn 'T3 tongue box-X 34x34' at the B-17 centre (Hypalon, coated face out).	T3 x2, B-17 x2	BT / PS (prototype WF2 + J1)	Nm 140/22 DPx17, round point R	BN tex135	7	301 box-X path / BOX	902	SK-1; QC-07; T3 (load table)	0.40 / 2.9
2.04	Fold the seam-inserted loops	Fold and clip each loop, raw ends together; no stitching here: the seam or bartack that inserts it sews it (ops named in its use). K-W x4 (80 mm): fold, raw ends in the seam at 40 mm; HPL x4 (80 mm): fold, raw ends in the seam at 40 mm; E5 x1 (60 mm): fold at 30 mm; I2T x1 (100 mm): fold at 50 mm; I2K x2 (60 mm): fold at 30 mm.	K-W x4, HPL x4, E5 x1, I2T x1, I2K x2	bench	-	-	-	- / -		SK-7	0.36 / 2.6
2.05	IME fairlead eyes (frame kit)	NECK FIRST (BL-044, as op 4.05's HE): on the FLAT strip fold each long edge 10 mm in between the two 'neck lines' (18 mm wide: the eye passes the 20 mm throat of the BL-045 fairlead) and sew one BT-M tack across each leg at the 'neck tack' lines [ZZ or BT; Nm 110/18 DPx17, round point R; bonded_nylon_tex90; ISO 304, 16 x 2 mm zigzag; prototype without zigzag: BT-301 fallback (4 passes, UNRATED)]. Then fold each 38 mm IME at 100 mm into an eye (neck folds INSIDE), box-X 34 x 30. These go on the frame fairleads, not the bag.	IME x6	BT / PS (prototype WF2 + J1)	Nm 140/22 DPx17, round point R	BN tex135	7	301 box-X path / BOX	1312	QC-07; IME	0.60 / 4.3
3.01	SH-07 slider garage on the right blade	Turn the edges 8 mm and topstitch SH-07 to B-02u at its placement.	SH-07, B-02u	WF1	Nm 110/18 DPx17, round point R	BN tex90	8	301 / LSd	200		0.16 / 1.1
3.02	IMP-4 onto the door, INSIDE face	Sew the IMP-4 bellows edge to B-02door at the IMP-4 placement (inside), pivot at the drill marks.	IMP-4, B-02door	WF1	Nm 90/14 DPx17, round point R	BN tex70	10	301 / SSa	820		0.69 / 5.0
3.03	Door #10 zip	Basting-tape the #10 zip to the door edge and to the blade U inner edge (1358 mm, 53.5 in), notch to notch, corner tangent notches at each R 55 corner; 2 sliders head to head; sew each tape at 5 mm, turn, edge-stitch 3 mm; zip stops at the hinge band.	B-02door, B-02u, #10 zip	WF1	Nm 110/18 DPx17, round point R	BN tex90	8	301 / ZIP	5432	SK-2; QC-09	2.66 / 19.1
3.04 V	B-22 guy anchors INSIDE the hinge band	Place each B-22 (coated face down on the band's inside) at x +/-130.3 mm (5.13 in) on the drawn placement (drawn x +/-149, moved 18.7 mm inboard, ASSUMPTION B22-IN). Double-row perimeter 3 / 9 mm.	B-22 x2, B-02h	WF2	Nm 110/18 DPx17, round point R	BN tex90	8	301 / LSd (2 rows)	839	SK-4; QC-07	0.67 / 4.8
3.05 V	Guy ladder-lock tabs LLT through B-22	Fold each LLT round the bar of a heavy 25 mm ladder lock, box-X through LLT + B-22 + B-02h at the drawn 'B-22 / guy LLT box' (drill corners). Box 21 x 25 mm (LLT piece).	LLT x2, B-22, B-02h, 2 heavy ladder locks	BT / PS (prototype WF2 + J1)	Nm 140/22 DPx17, round point R	BN tex135	7	301 box-X path / BOX	315	SK-1, SK-4; QC-07; LLT + B-22 + B-02h	0.20 / 1.4
3.06	B-28 front tabs on the hinge band (OUTSIDE)	Fold a B-28t round the bar of a heavy ladder lock (lock bar on the top line of the drawn 'B-28 front lock tab' placement), box-X it through tab + band on the OUTSIDE of B-02h at the drawn 'B-28 front tab (OUTSIDE)' box, 21 x 20 mm. DECISION B28-FACE (BL-034): the crown strap runs down the band's aft face to the lock, so the tab is outside.	B-28t x2, B-02h, 2 heavy ladder locks	BT / PS (prototype WF2 + J1)	Nm 140/22 DPx17, round point R	BN tex135	7	301 box-X path / BOX	280	SK-4; QC-07; B-28t	0.20 / 1.4

Seq	Operation	How (beginner)	Pieces	Machine	Needle	Thread	SPI	Stitch / seam	mm	Ref	h / SAM min
3.07	Hinge band to the door top	Join B-02h to the door top (334 mm), SSa2.	B-02h, B-02door	WF1	Nm 110/18 DPx17, round point R	BN tex90	8	301 / SSa2	668		0.20 / 1.4
3.08	Hinge band ends to the blade U	Join both short band ends to the blade tops (193 mm total). The B-22 tabs sit 6 mm inside this seam line: turn the allowances AWAY from B-22 (onto the blade) and sew the SSa2 second row on the blade side, so it does not run onto the Hypalon edge.	B-02h, B-02u	WF1	Nm 110/18 DPx17, round point R	BN tex90	8	301 / SSa2	385		0.22 / 1.6
4.01 V	Check the B-01 cut-outs	6 swell windows, 2 B-27 slits (156 mm at x +/-52), IMP-2 keeper drill marks. All cut, corners awled.	B-01	bench	-	-	-	- / -		QC-03	0.10 / 0.7
4.02 V	Set the 6 swells into the B-01 windows	Swell base to window edge, INSIDE B-01, notch to notch (6 windows, 1680 mm of window edge in all; hinge windows: 0). Sew from the swell side, needle down at every corner.	B-13x x6, B-01	WF1	Nm 110/18 DPx17, round point R	BN tex90	8	301 / SSa	1680	SK-3; QC-08	1.50 / 10.8
4.03	B-27 guy ports (2)	Each slit (156 mm, x +/-52, z 595-745): turn both edges 6 mm and edge-stitch; sew the B-27f facing INSIDE along the inboard stitch line so the outboard edge laps 20 mm over it; bartack both slit ends. It crosses the setback crease at z 631: the facing follows the crease.	B-01, B-27f x2	WF1	Nm 110/18 DPx17, round point R	BN tex90	8	301 / EFa + LSd (lapped port)	965	SK-2, SK-3; QC-10; B-27 slit ends	0.98 / 7.0
4.04	B-17 IM3 tabs OUTSIDE B-01	Place each B-17 (with its T3) OUTSIDE B-01 at x +/-115, z 385-455 (placement). Double-row perimeter 3 / 9 mm, 8 SPI, tex 90; the 90 mm top edge carries the load: no gaps, backtack 10 mm.	B-17 x2, B-01	WF2	Nm 110/18 DPx17, round point R	BN tex90	8	301 / LSd (2 rows)	1239	SK-3; QC-07; B-17 perimeter	0.99 / 7.1
4.05 V	B-18 doublers + HE hinge eyes (DECISION HE-ONE / HE-PIN, sketch SK-8)	(1) NECK, on the FLAT HE strip (96 mm): fold each long edge 10 mm in between the two 'neck lines' (the strip there is 18 mm wide, the fairlead throat): fold it ROUND an 18.0 mm gauge bar (DECISION HE-NECK-GAUGE, BL-048: the webbing's width tolerance goes into the fold, not the neck), then one BT-M bartack across each leg at the 'neck tack' lines (through that leg only) [ZZ or BT; Nm 110/18 DPx17, round point R; bonded_nylon_tex90; ISO 304, 16 x 2 mm zigzag; prototype without zigzag: BT-301 fallback (4 passes, UNRATED)]; every neck must pass a 18.6 mm go / no-go slot (QC; the same gauge for the IME and US necks). (2) B-18 ROWS FIRST: lay B-18 INSIDE B-01 on the drawn placement, coated face to B-01, its lower edge ON the B-01 cut edge (it runs 12 mm below the floor seam line); sew its perimeter rows 3 and 9 mm in, on the TOP and BOTH SIDES only (the bottom goes in the floor seam), through B-18 + B-01 ONLY [WF2; Nm 110/18 DPx17, round point R; bonded_nylon_tex90; ISO 301, 8 SPI; rated rows (load_points B-18)]. Nothing else is on the panel yet: these rows must never catch the HE. (3) PLACE THE HE: fold the strip in half on the 'EYE FOLD' line (raw ends together, neck folds inside) and lay it OUTSIDE B-01 over B-18, centred on the 'hinge eye CL' notch, eye fold pointing UP, both raw ends ON the B-01 cut edge. Layer order from outside: HE leg / HE leg / B-01 / B-18. Clip. (4) BOX: box-X 34 x 7 at the drawn 'HE hinge eye' box (drill corners: its lower edge 4 mm, its top 11 mm above the floor seam line), through all four layers [BT / PS (prototype: WF2 + J1 template); Nm 140/22 DPx17, round point R; bonded_nylon_tex135; ISO 301, box-X path 7 SPI]. Its top edge must stay BELOW the fairlead cheeks: nothing but the 18 mm neck may sit above +16 mm. (5) BASTE the bottom of the stack (2 HE raw ends + B-18 + B-01) at 6 mm, inside the allowance, so op 4.15 catches all of it [WF1; Nm 110/18 DPx17, round point R; bonded_nylon_tex90; ISO 301, 6 mm stitch]. The free loop above the box (25 mm, fold 36 mm above the seam line) is the eye: it points UP and takes the z 60 pin with the 18 mm neck in the 20 mm throat and the 3-ply neck through the 6.5 mm gap under the pin (DECISION HE-ENV, BL-042 hinge fairlead; loaded pin centre 25.1 mm above the seam line, HE-FIT band 22-29). No stitching crosses the eye after step 3.	HE x2, B-18 x2, B-01	BT / PS (prototype WF2 + J1)	Nm 140/22 DPx17, round point R	BN tex135	7	301 box-X path / LSd + BOX	1223	SK-8, SK-1, SK-3; QC-07; HE + B-18 + B-01	1.21 / 8.7
4.06	IMP-2 keeper tabs I2K on B-01	Fold each I2K, bartack at the 2 'IMP-2 keeper tab' drill marks (inside).	I2K x2, B-01	BT	Nm 110/18 DPx17, round point R	BN tex90	-	304 / BT			0.20 / 1.4

Seq	Operation	How (beginner)	Pieces	Machine	Needle	Thread	SPI	Stitch / seam	mm	Ref	h / SAM min
4.07	B-11 lash tabs (8) on the gussets	Gusset flat work, before the gusset seams: 4 tabs on each B-03 at the drawn 'B-11 lash tab' placements (bow riser / lash z 470 (175 mm aft of the frame seam, 40 mm forward of the aft corner); bow riser / lash z 560 (186 mm aft of the frame seam, 40 mm forward of the aft corner); muzzle keeper z 650 (113 mm aft of the frame seam); muzzle keeper z 650 (173 mm aft of the frame seam)). Tab coated face out, slot end UP; awl the 4 'B-11 box corner' drill marks; sew the 24 x 19 mm 'B-11 sew-down box' (rectangle + both diagonals, 301) through tab + gusset. Right gusset: bow riser; left: LIGHT LASH ONLY, no load rating (BL-040 m-12); muzzle keepers both sides, AFT of the roll-clip path (BL-041 RT-FWD: 10 mm clear of the buckle; pattern ASSUMPTION B11-PLACE / MK-GAP, settled by the archery and rifle loadout fits and FIT-R3).	B-11 x8, B-03 x2	WF2	Nm 110/18 DPx17, round point R	BN tex90	8	301 / LSd (box-X sew-down)	1178	SK-5; QC-07	0.80 / 5.8
4.08	RTM roll-clip buckle tabs on the gussets	Gusset flat work, before the gusset seams. (0) BEFORE the placement is chalked (BL-048 m-3): mate one bought 25 mm side-release buckle and caliper it: width, and height from the female bar to the far end of the mated male. The placement assumes 35 wide x 35 above the bar (ASSUMPTION BUCKLE-ENV; typical mated 25 mm buckles stand 55-65 mm, fourth review m-3). If either reading is larger, STOP: send both readings to the pack-designer, who re-runs patterns_corrie_f_rev3.rt_layout with them (BUCKLE_ENV) and re-issues the B-03 placement; at kit S 15 in W3's lowest edge must stay >= 10 mm clear of the mated male half. (1) Thread each RTM through the bar of a side-release buckle's FEMALE half (the half with the socket) and fold it on its fold line. (2) Lay it on the OUTSIDE of each B-03 on the drawn 'RTM roll-clip buckle tab' placement (z 544, 202 mm under the collar seam, 60 mm aft of the frame seam line): buckle bar on the top line, socket pointing UP (the male half on the RT strap comes down into it from the roll end along the drawn 'RT strap from the roll end' line). DECISION W3-OVER (BL-043): the buckle sits 10 mm under the lowest W3 edge of every crown kit (W3 z 601-804 with the torso setting), so W3 never bears on it; nothing else is sewn on the strap path (the BL-041 W3G guide is deleted). (3) Box-X 21 x 25 at the drawn 'RTM buckle tab' box through RTM (2 plies) + B-03 [BT / PS (prototype: WF2 + J1 template); Nm 140/22 DPx17, round point R; bonded_nylon_tex135; ISO 301, box-X path 7 SPI].	RTM x2, B-03 x2, 2 SR buckle female halves	BT / PS (prototype WF2 + J1)	Nm 140/22 DPx17, round point R	BN tex135	7	301 box-X path / BOX	315	SK-10; QC-07; RTM	0.30 / 2.2
4.09	Gusset flat work: kick bands	SH-01b on each gusset foot at the 'kick band top' line, top edge turned 8 mm, topstitched; bottom and sides go in the seams.	SH-01b x2, B-03 x2	WF1	Nm 110/18 DPx17, round point R	BN tex90	8	301 / LSd	320		0.26 / 1.9
4.10 V	Gusset flat work: FP pockets (both)	Sew each FP to its gusset at the placement, z 150-380, 165 mm face; inner edge >= 9 mm off the frame seam (9.5 mm at the tightest point, pattern fp_min_margin_mm). Bottom and FRAME-side edge 301 at 3 mm, box pleats laid flat, bartack the 2 mouth corners. The AFT edge lies on the aft corner seam line: baste it at 6 mm inside the allowance only; the aft corner seam (op 4.13) sews it, UNDER the K-W keeper loops (SK-9).	FP x2, B-03 x2	WF1	Nm 90/14 DPx17, medium ball SUK (knit mesh)	BN tex70	10	301 / LSd	1250	SK-5; FP corners (BT-S)	1.15 / 8.3
4.11	Gusset flat work: IMP-8 (R only) and hose ports	IMP-8 cup at 'IMP-8 cup top (R only)'. Hose slots (both gussets, z 558; BL-043 DECISION HOSE-LOW: below W3 at every crown setting, so a cinched W3 never lies on the hose): B-09p binding (TPU, laser-cut) laid over the slot, perimeter + slot edge stitched.	IMP-8, B-09p x2, B-03 x2	WF1	Nm 110/18 DPx17, round point R	BN tex90	8	301 / LSd	1090		0.81 / 5.9
4.12	Frame-side box seams B-01 | gussets	R (720 mm): E5 pole-grip tab at z 640 and the ONE end of the B-26 shock cord inserted 15 mm at the 'B-26 tripod keeper' notch (the other end stays loose until op 4.13). L (720 mm): 2 HPL hard-point loops at z 420 / 640 inserted. Notch to notch every 50 mm; SSa2.	B-01, B-03 x2, E5, E5-cord, B-26, HPL x2	WF1	Nm 110/18 DPx17, round point R	BN tex90	8	301 / SSa-ins + SSa2	2881	SK-5; QC-08	1.16 / 8.4

Seq	Operation	How (beginner)	Pieces	Machine	Needle	Thread	SPI	Stitch / seam	mm	Ref	h / SAM min
4.13	Aft corner seams blade U | gussets (closes the tube)	R and L (715 / 715 mm): 2 HPL loops in the L seam at the hard-point notches; 2 K-W keeper loops in each seam, raw ends in the seam, at the W1-W3 crossings: R seam z 128 / 300; L seam z 202 / 370 (W1, W2; the drawn 'K-W keeper' notches on B-03 and B-02u, pattern ASSUMPTION KW-PATH). R seam: the B-26 free end inserted 15 mm at the 'B-26 keeper' notch. SSa2. LAYER ORDER where a K-W keeper meets the FP pocket edge (SK-9; it happens at L z 202 (W1), R z 300 (W2), L z 370 (W2)): gusset B-03 face up, FP mesh edge on it (already basted), the K-W loop ON TOP of the mesh (raw ends at the cut edge, loop pointing into the gusset), then the blade U face down. Sew. Turned out, the keeper sits OUTSIDE the mesh, so the haul wrap runs over the pocket, not inside it. Check each one before sewing the next 100 mm: a keeper under the mesh means unpicking this 715 mm seam.	B-02u, B-03 x2, HPL x2, K-W x4, B-26	WF1	Nm 110/18 DPx17, round point R	BN tex90	8	301 / SSa-ins + SSa2	2861	SK-9, SK-5; QC-08	1.66 / 11.9
4.14 V	US floor channels B-16 (US1) and B-16b (US2) on F-02	Turn both ends of B-16 and of B-16b 15 mm under (left OPEN: the 25 mm under-strap passes through). B-16 (US1, 140 mm): lay it on F-02's outer face on the drawn 'B-16 channel US1' lines, its ends on the 'channel end (open)' marks (x +/-70), topstitch both long edges, bartack the 4 corners. B-16b (US2, 330 mm, ends at x +/-165): centre it on F-02's 'B-16b channel US2' lines (CL notches matched), topstitch both long edges ON F-02 ONLY, from F-02's left edge to its right edge; leave the two 70 mm tails FREE (they are sewn to B-05 in op 'Floor B-05'). Stations from the floor's frame edge 48 / 104 mm. Do not swap the two: B-16 is the short one.	B-16, B-16b, F-02	WF1	Nm 110/18 DPx17, round point R	BN tex90	8	301 / LSd	660		0.73 / 5.2
4.15	Floor B-05 (F-02 first)	Sew F-02 to the OUTSIDE of B-05 on the drawn 'F-02 fly pocket' placement (zip edge aft, corner drill marks), perimeter row. Then the floor ring (1107 mm): B-01 edge first with the HE tails and the B-18 lower edges caught in it, then gussets, then the blade kick strip; quarter notches matched, needle down at the 4 corners. SSa2. At the two hinge eyes turn the allowances DOWN onto B-05 (the second row runs on the floor, clear of the HE box, ASSUMPTION HE-GAP). BL-041 n-14: sew the two hinge-eye spans (x +/-115, 50 mm each way) with the ZIPPER / EDGE FOOT: the floor seam runs 4 mm from the HE box stack and a walking foot (5-6 mm half width) rides up on it there; check from inside that the B-18 lower edge and both HE raw ends are in the seam. BL-050 B: after F-02 is on and BEFORE the floor ring, lay B-16b's two tails flat on B-05 on the drawn 'B-16b US2 channel tail edge' lines (ends on the 'channel end (open)' marks at x +/-165, 32 mm inside the side seam line), topstitch both long edges through B-05 and bartack the 2 outer corners of each tail; then seal those stitch lines (and the bartacks) on B-05's INSIDE face with the urethane seam sealer (BOM consumable), a thin bead pressed into the holes, 12 h to cure before the floor ring is folded.	B-05, F-02, B-16b tails, B-01, B-03 x2, B-02u	WF1	Nm 110/18 DPx17, round point R	BN tex90	8	301 / SSa2	3154	QC-08	2.52 / 18.1
5.01	Collar tube	B-06f (ULTRA) to B-06 both sides (990 mm); I2T hang tab at the B-06f drill mark.	B-06f, B-06, I2T	WF1	Nm 90/14 DPx17, round point R	BN tex70	10	301 / SSa2	1979		0.69 / 5.0
5.02	RT roll-clip straps on the flat hem B-07	B-07 still flat. Each RT strap (565 mm): lay its first 60 mm (to the 'end of the sewn zone' line) on the OUTSIDE of B-07 on the drawn 'RT roll-clip strap' rectangle (L at x 967-1027 mm, R at x 382-442 mm from the B-07 left end), strap end on the outer line, the free strap running along the hem out across the 'roll end' notch (BL-043 DECISION RT-ORIGIN: the strap leaves the roll at its end, where the hem folds; its box sits on the B-07s strip, 7.5 mm clear of the 40 mm fold gap). Box-X 21 x 25 at the drawn 'RT strap' box through strap + B-07. Fold the free strap back over the box for now and clip it out of the way. The ADJUSTABLE (male) buckle half is threaded on the free end at hardware fitting (op 7.25).	RT x2, B-07	BT / PS (prototype WF2 + J1)	Nm 140/22 DPx17, round point R	BN tex135	7	301 box-X path / BOX	315	SK-10; QC-07; RT	0.30 / 2.2

Seq	Operation	How (beginner)	Pieces	Machine	Needle	Thread	SPI	Stitch / seam	mm	Ref	h / SAM min
5.03	Roll-top hem with HDPE strips	Close the hem back seam (70 mm). Sew the hem to the collar top ring (1050 mm), the RT straps hanging free outside; fold over the two B-07s HDPE strips, their ends on the drawn 'B-07s strip end' lines (a 40 mm gap at each roll end, where the hem folds flat), and stitch closed (EFb), keeping the strap ends out of the fold.	B-07, B-07s x2, B-06	WF1	Nm 110/18 DPx17, round point R	BN tex90	8	301 / SSa + EFb	2169	SK-2	1.17 / 8.4
5.04	Collar ring to the body	Collar to the body tops (1050 mm), matched at every panel-corner V-notch. The seam closes the top end of each B-27 port; bartack the port top over it.	collar, body	WF1	Nm 110/18 DPx17, round point R	BN tex90	8	301 / SSa2	2100	QC-08, QC-10	1.33 / 9.6
6.01	Lid: 4 crown darts sewn + pressed	Fold each B-29a corner dart on its centre, sew leg to leg from the edge to the apex drill mark, tie off (no backtack at the apex), press the dart toward the aft edge. (rows from patterns_corrie_f_rev3.lid_hours)	B-29a, B-29b, B-29c, B-29d, B-29e, B-29f, B-29g	WF1	Nm 110/18 DPx17, round point R	BN tex90	8	301 / DART		SK-6	1.00 / 7.2
6.02	Lid: strap channels topstitched to the crown, both edges (2 x 2 runs)	Lay each B-29g on its 'strap channel CL' line (x +/-52), edges turned 8 mm, topstitch both long edges; channels are 30 mm wide for the 25 mm B-28 strap (drawn so). (rows from patterns_corrie_f_rev3.lid_hours)	B-29a, B-29b, B-29c, B-29d, B-29e, B-29f, B-29g	WF1	Nm 110/18 DPx17, round point R	BN tex90	8	301 / LSd	1062	SK-6	0.64 / 4.6
6.03	Lid: bartacks at the 4 channel ends	BT-M bartack across each channel end (tex 90, 304), 3 mm in from the turned end. (rows from patterns_corrie_f_rev3.lid_hours)	B-29a, B-29b, B-29c, B-29d, B-29e, B-29f, B-29g	BT	Nm 110/18 DPx17, round point R	BN tex90	-	304 / BT		SK-6	0.40 / 2.9
6.04	Lid: camo caps: short ends turned and topstitched (2 x 2)	Turn the short ends of each B-29f 8 mm (its pattern allowance) and topstitch the cap on B-29e at its placement; long edges go into the ring seams. (rows from patterns_corrie_f_rev3.lid_hours)	B-29a, B-29b, B-29c, B-29d, B-29e, B-29f, B-29g	WF1	Nm 110/18 DPx17, round point R	BN tex90	8	301 / LSd	60	SK-6	0.05 / 0.4
6.05	Lid: #5 coil zip: tape baste, stops, slider	Baste the #5 zip between B-29d (upper) and B-29e (lower) with 6 mm tape; zip stops at the drill marks; slider on before closing. (rows from patterns_corrie_f_rev3.lid_hours)	B-29a, B-29b, B-29c, B-29d, B-29e, B-29f, B-29g	WF1	Nm 110/18 DPx17, round point R	BN tex90	8	301 / ZIP		SK-6	0.50 / 3.6
6.06	Lid: #5 coil zip sewn, both tapes	ZIP seam: each tape at 5 mm, turn, edge-stitch 3 mm. (rows from patterns_corrie_f_rev3.lid_hours)	B-29a, B-29b, B-29c, B-29d, B-29e, B-29f, B-29g	WF1	Nm 110/18 DPx17, round point R	BN tex90	8	301 / ZIP	940	SK-6	0.47 / 3.4
6.07	Lid: zip round the two aft corners	Ease the zip round each aft corner; clip the tape 3 mm at the corner tangent notches. (rows from patterns_corrie_f_rev3.lid_hours)	B-29a, B-29b, B-29c, B-29d, B-29e, B-29f, B-29g	WF1	Nm 110/18 DPx17, round point R	BN tex90	8	301 / ZIP		SK-6	0.80 / 5.8
6.08	Lid: wall ring closed: two short seams	Join the front wall run B-29c to the aft strips at both ends (30 mm seams), SSa2. (rows from patterns_corrie_f_rev3.lid_hours)	B-29a, B-29b, B-29c, B-29d, B-29e, B-29f, B-29g	WF1	Nm 110/18 DPx17, round point R	BN tex90	8	301 / SSa2	120	SK-6	0.04 / 0.3
6.09	Lid: crown | wall perimeter (curved, notch match, bind)	Crown to wall ring notch to notch, darts matched to the corner notches; clip the crown allowance every 15 mm on the curves; crown on top. (rows from patterns_corrie_f_rev3.lid_hours)	B-29a, B-29b, B-29c, B-29d, B-29e, B-29f, B-29g	WF1	Nm 110/18 DPx17, round point R	BN tex90	8	301 / SSa2	1978	SK-6	1.78 / 12.8
6.10	Lid: crown | wall: closed ring handling	Sew the ring in one pass, needle down at each notch; open the zip first. (rows from patterns_corrie_f_rev3.lid_hours)	B-29a, B-29b, B-29c, B-29d, B-29e, B-29f, B-29g	WF1	Nm 110/18 DPx17, round point R	BN tex90	8	301 / SSa2		SK-6	0.50 / 3.6
6.11	Lid: wall | base perimeter (curved, notch match, bind), turned through the zip	Base B-29b to the wall ring, notch to notch, inside out; then turn the lid right side out through the open zip. (rows from patterns_corrie_f_rev3.lid_hours)	B-29a, B-29b, B-29c, B-29d, B-29e, B-29f, B-29g	WF1	Nm 90/14 DPx17, round point R	BN tex70	10	301 / SSa2	1978	SK-6	1.78 / 12.8
6.12	Lid: wall | base: closed ring handling	As the crown ring: one pass, needle down at each notch. (rows from patterns_corrie_f_rev3.lid_hours)	B-29a, B-29b, B-29c, B-29d, B-29e, B-29f, B-29g	WF1	Nm 90/14 DPx17, round point R	BN tex70	10	301 / SSa2		SK-6	0.50 / 3.6
7.01	B-28: B-28 crown strap, kit M x2	straight strap, sealed ends; threaded at assembly (op 7.25).	B-28 x2	bench	-	-	-	- / -		SK-1, SK-7	0.00 / 0.0
7.02	B-28t: lock tab (the other 2 are boxed in an earlier op) x2	60 mm (2.4 in). Folds: fold round the lock bar. Box-X: 21 x 20 at the drawn marks.	B-28t x2	BT / PS (prototype WF2 + J1)	Nm 140/22 DPx17, round point R	BN tex135	7	301 box-X path / BOX	280	SK-1, SK-7; QC-07; B-28t	0.20 / 1.4

Seq	Operation	How (beginner)	Pieces	Machine	Needle	Thread	SPI	Stitch / seam	mm	Ref	h / SAM min
7.03	B-28s: toggle loop x2	60 mm (2.4 in). Folds: fold: toggle loop. Box-X: 21 x 15 at the drawn marks.	B-28s x2	BT / PS (prototype WF2 + J1)	Nm 140/22 DPx17, round point R	BN tex135	7	301 box-X path / BOX	247	SK-1, SK-7; QC-07; B-28s	0.20 / 1.4
7.04	K-28: B-28 tail keeper (25 mm elastic loop) x4	Cut 90 mm, lap 15 mm into a loop round its strap, zigzag the lap (3 passes).	K-28 x4	ZZ	Nm 110/18 DPx17, round point R	BN tex90	8	304 / BT (zigzag lap)			0.40 / 2.9
7.05	W1: haul wrap W1 (cut for carry) x1	750 mm (29.5 in). Folds: G-hook fold. Box-X: 21 x 25 at the drawn marks.	W1 x1	BT / PS (prototype WF2 + J1)	Nm 140/22 DPx17, round point R	BN tex135	7	301 box-X path / BOX	157	SK-1, SK-7; QC-07; W1	0.10 / 0.7
7.06	W2: haul wrap W2 (cut for carry) x1	822 mm (32.4 in). Folds: G-hook fold. Box-X: 21 x 25 at the drawn marks.	W2 x1	BT / PS (prototype WF2 + J1)	Nm 140/22 DPx17, round point R	BN tex135	7	301 box-X path / BOX	157	SK-1, SK-7; QC-07; W2	0.10 / 0.7
7.07	W3: haul wrap W3, kit M x1	987 mm (38.8 in). Folds: G-hook fold. Box-X: 21 x 25 at the drawn marks.	W3 x1	BT / PS (prototype WF2 + J1)	Nm 140/22 DPx17, round point R	BN tex135	7	301 box-X path / BOX	157	SK-1, SK-7; QC-07; W3	0.10 / 0.7
7.08	WX: W haul extender (meat-trip kit) x3	740 mm (29.1 in). Folds: G-hook fold; sewn loop fold (takes the W's own G-hook). Box-X: 21 x 25, 21 x 25 at the drawn marks.	WX x3	BT / PS (prototype WF2 + J1)	Nm 140/22 DPx17, round point R	BN tex135	7	301 box-X path / BOX	944	SK-1, SK-7; QC-07; WX	0.60 / 4.3
7.09	K-RT: roll clip tail keeper (25 mm elastic) x2	Cut 90 mm, lap 15 mm into a loop round its strap, zigzag the lap (3 passes).	K-RT x2	ZZ	Nm 110/18 DPx17, round point R	BN tex90	8	304 / BT (zigzag lap)			0.20 / 1.4
7.10	IM1: internal meat strap IM1 x1	990 mm (39.0 in). Folds: left end loop (girth-hitched on the eye). Box-X: 21 x 25 at the drawn marks.	IM1 x1	BT / PS (prototype WF2 + J1)	Nm 140/22 DPx17, round point R	BN tex135	7	301 box-X path / BOX	157	SK-1, SK-7; QC-07; IM1	0.10 / 0.7
7.11	IM2: internal meat strap IM2 x1	1023 mm (40.3 in). Folds: left end loop (girth-hitched on the eye). Box-X: 21 x 25 at the drawn marks.	IM2 x1	BT / PS (prototype WF2 + J1)	Nm 140/22 DPx17, round point R	BN tex135	7	301 box-X path / BOX	157	SK-1, SK-7; QC-07; IM2	0.10 / 0.7
7.12	IM3: internal meat strap IM3 x1	1083 mm (42.6 in). Folds: left end loop (girth-hitched on the eye). Box-X: 21 x 25 at the drawn marks.	IM3 x1	BT / PS (prototype WF2 + J1)	Nm 140/22 DPx17, round point R	BN tex135	7	301 box-X path / BOX	157	SK-1, SK-7; QC-07; IM3	0.10 / 0.7
7.13	IMC: IM cam anchor tab (cam bar <-> hook / carabiner) x3	HARDWARE FIRST (BL-043 m-5): this op closes both loops, so before sewing either box thread the 25 mm CAM's anchor bar into the 'fold round the CAM anchor bar' loop (all 3 tabs) and the rated G-HOOK's bar into the other loop of the IM1 and IM2 tabs; the IM3 tab's second loop stays EMPTY (the wiregate carabiner clips in at op 7.25). Hardware for this op: 3 cams, 2 G-hooks. Then: 192 mm (7.6 in). Folds: fold round the CAM anchor bar; fold round the G-hook bar / carabiner. Box-X: 21 x 25, 21 x 25 at the drawn marks.	IMC x3, 3 x 25 mm cam, 2 x rated G-hook	BT / PS (prototype WF2 + J1)	Nm 140/22 DPx17, round point R	BN tex135	7	301 box-X path / BOX	944	SK-1, SK-7; QC-07; IMC	0.60 / 4.3
7.14	K-IM: IM tail keeper (25 mm elastic) x3	Cut 90 mm, lap 15 mm into a loop round its strap, zigzag the lap (3 passes).	K-IM x3	ZZ	Nm 110/18 DPx17, round point R	BN tex90	8	304 / BT (zigzag lap)			0.30 / 2.2
7.15 V	US: under-strap US1 (pins z 95) x1	NECK FIRST (BL-044, as op 4.05's HE): on the FLAT strip fold each long edge 3.5 mm in between the two 'neck lines' (18 mm: the eye passes the 20 mm throat of the BL-045 fairlead), one BT-M tack across each leg at the 'neck tack' lines [ZZ or BT; Nm 110/18 DPx17, round point R; bonded_nylon_tex90; ISO 304, 16 x 2 mm zigzag; prototype without zigzag: BT-301 fallback (4 passes, UNRATED)]. 550 mm (21.7 in). Folds: eye fold (round the left pin); neck: fold the edge in to this line; neck: fold the edge in to this line. Box-X: 21 x 25 at the drawn marks.	US x1	BT / PS (prototype WF2 + J1)	Nm 140/22 DPx17, round point R	BN tex135	7	301 box-X path / BOX	157	SK-1, SK-7; QC-07; US	0.10 / 0.7

Seq	Operation	How (beginner)	Pieces	Machine	Needle	Thread	SPI	Stitch / seam	mm	Ref	h / SAM min
7.16 V	US2: under-strap US2 (pins z 142) x1	NECK FIRST (BL-044, as op 4.05's HE): on the FLAT strip fold each long edge 3.5 mm in between the two 'neck lines' (18 mm: the eye passes the 20 mm throat of the BL-045 fairlead), one BT-M tack across each leg at the 'neck tack' lines [ZZ or BT; Nm 110/18 DPx17, round point R; bonded_nylon_tex90; ISO 304, 16 x 2 mm zigzag; prototype without zigzag: BT-301 fallback (4 passes, UNRATED)]. 925 mm (36.4 in). Folds: eye fold (round the left pin); neck: fold the edge in to this line; neck: fold the edge in to this line. Box-X: 21 x 25 at the drawn marks.	US2 x1	BT / PS (prototype WF2 + J1)	Nm 140/22 DPx17, round point R	BN tex135	7	301 box-X path / BOX	157	SK-1, SK-7; QC-07; US2	0.10 / 0.7
7.17	USA: US cam anchor (right pin <-> cam bar) x2	NECK FIRST (BL-044, as op 4.05's HE): on the FLAT strip fold each long edge 3.5 mm in between the two 'neck lines' (18 mm: the eye passes the 20 mm throat of the BL-045 fairlead), one BT-M tack across each leg at the 'neck tack' lines [ZZ or BT; Nm 110/18 DPx17, round point R; bonded_nylon_tex90; ISO 304, 16 x 2 mm zigzag; prototype without zigzag: BT-301 fallback (4 passes, UNRATED)]. HARDWARE FIRST: thread the US CAM's anchor bar into the 'fold round the CAM anchor bar' loop before its box is sewn; the necked 'RIGHT PIN' loop (23 mm clear, DECISION USA-LOOP) is sewn closed EMPTY (no pin in it: at frame-kit assembly the pin goes through the cheek sleeve, the loop and the other cheek sleeve). Hardware for this op: 2 cams. Then: 202 mm (8.0 in). Folds: fold round the RIGHT PIN; fold round the CAM anchor bar; neck: fold the edge in to this line; neck: fold the edge in to this line. Box-X: 21 x 25, 21 x 25 at the drawn marks.	USA x2	BT / PS (prototype WF2 + J1)	Nm 140/22 DPx17, round point R	BN tex135	7	301 box-X path / BOX	629	SK-1, SK-7; QC-07; USA	0.40 / 2.9
7.18	B-24: float stand-off link (38 mm, meat-trip) x2	435 mm (17.1 in). Folds: fold round the hook bed (finished 0); loop 1 start; loop 1 apex: gap 120 mm (4.7 in) - T3 hook sits here; loop 1 end (meets start); loop 2 start; loop 2 apex: gap 170 mm (6.7 in) - T3 hook sits here; loop 2 end (meets start); loop 3 start; loop 3 apex: gap 220 mm (8.7 in) - T3 hook sits here; loop 3 end (meets start). Box-X: 34 x 30, 34 x 20, 34 x 20, 34 x 20 at the drawn marks.	B-24 x2	BT / PS (prototype WF2 + J1)	Nm 140/22 DPx17, round point R	BN tex135	7	301 box-X path / BOX	1559	SK-1, SK-7; QC-07; B-24	0.80 / 5.8

Seq	Operation	How (beginner)	Pieces	Machine	Needle	Thread	SPI	Stitch / seam	mm	Ref	h / SAM min
7.19	B-25: guy bridle on the crown top bar (38 mm, BL-041 / BL-043 topology) x1	513 mm (20.2 in). Folds: tab fold R (toggle loop); bar sleeve R start; bar sleeve R lap (bartack); leaves the bar DOWN, aft side; spine wrap centre (round the spine); bar sleeve L lap (bartack); bar sleeve L start; tab fold L (toggle loop); MITRE R: 45 deg fold, turns the strip 90 deg inboard (under the sleeve); MITRE L: 45 deg fold, turns the strip 90 deg inboard (under the sleeve); NECK: fold the lower 13 mm (x 0 edge = lower edge) up between the two band-edge marks. Box-X: 21 x 15, 21 x 15 at the drawn marks. Toggle tabs: fold each end on its 'tab fold' line and sew the drawn tip-end box (DECISION B25-BOX: the clear loop passes the POM toggle). TOPOLOGY (BL-043 DECISION B25-TOPO, sketch SK-12): round the bar = the sleeve; the strip leaves the bar DOWN on its aft (bag) side inside the sleeve band, drops 27 mm, turns 90 deg inboard at the drawn 45 deg 'MITRE' fold (under the sleeve, x 26.5-64.5), is NECKED to 25 mm on the 'NECK' line (lower edge folded up) from the band edge inboard, and wraps once round the spine under the clamp collar, continuous to the other side. Nothing but the spine wrap is on the wearer side of the bar (the run reaches the spine fore of the bar plane, so the wrap goes round it there; the sleeves, drops and mitres are all aft). NECK FOLD: secure each 13 mm neck fold with one BT-M tack [ZZ or BT; Nm 110/18 DPx17, round point R; bonded_nylon_tex90; ISO 304, 16 x 2 mm zigzag; prototype without zigzag: BT-301 fallback (4 passes, UNRATED)] 10 mm inboard of the band edge, and one BT-M across the mitre fold, both sewn flat now (BL-048 m-5). CUT LAST (BL-048 m-4): the spine wrap is booked as 2 pi r but the runs arrive at the spine at an angle in plan, so the cut is uncertain by about +/-10-18 mm: cut B-25 after every other webbing part, from a paper or scrap-strip dry fit at B25-FIT, and mark the cut from the strip. FRAME COORDINATION: the bar sleeves close round the crown top bar BEFORE the crown corners are bonded (H1 build). Sew each sleeve lap OFF the bar, round an 18 mm dowel standing in for the bar (BT-M [ZZ or BT; Nm 110/18 DPx17, round point R; bonded_nylon_tex90; ISO 304, 16 x 2 mm zigzag; prototype without zigzag: BT-301 fallback (4 passes, UNRATED)]), then slide the closed sleeve onto the bar from its free end before the corners are bonded. GATE B25-FIT (before the laps are sewn): dry-fit the bridle (or the scrap strip) on the crown with the clamp fitted and the crown at kit S's SHORTEST setting (15 in): the mitres sit inside the sleeve bands, the necked run passes under the clamp collar, and the spine wrap clears the pawl housing (design 23 mm on the BL-046 clamp constants and the measured housing; the run reaches the spine 25.02 mm FORE of the bar plane); a no stops the laps: photo to the pack-designer.	B-25 x1	BT / PS (prototype WF2 + J1)	Nm 140/22 DPx17, round point R	BN tex135	7	301 box-X path / BOX	247	SK-1, SK-7, SK-12; QC-07; B-25	0.20 / 1.4
7.20	TG: port guy (B-25 toggle -> B-27 port -> B-22 inside) x2	500 mm (19.7 in). Folds: -. Box-X: 21 x 25 at the drawn marks.	TG x2	BT / PS (prototype WF2 + J1)	Nm 140/22 DPx17, round point R	BN tex135	7	301 box-X path / BOX	315	SK-1, SK-7; QC-07; TG	0.20 / 1.4
7.21	K-G: guy tail keeper (25 mm elastic) x2	Cut 90 mm, lap 15 mm into a loop round its strap, zigzag the lap (3 passes).	K-G x2	ZZ	Nm 110/18 DPx17, round point R	BN tex90	8	304 / BT (zigzag lap)			0.20 / 1.4
7.22	BRL: bow riser loop (through B-11) x2	100 mm (3.9 in). Folds: fold. Box-X / bartack per the load table where it is caught in a seam.	BRL x2	BT	Nm 110/18 DPx17, round point R	BN tex90	-	304 / BT		SK-1, SK-7	0.20 / 1.4
7.23	BRS: bow riser strap + G-hook x1	300 mm (11.8 in). Folds: G-hook fold. Box-X / bartack per the load table where it is caught in a seam.	BRS x1	BT	Nm 110/18 DPx17, round point R	BN tex90	-	304 / BT		SK-1, SK-7	0.10 / 0.7
7.24	MK-cord: rifle muzzle keeper, 3 mm shock cord loop through the 2 B-11 muzzle tabs + spring cord lock (one per gusset) x2	Cut 299 mm (11.8 in), melt / whip both ends; fitted at hardware assembly (op below) as its use says.	MK-cord x2	bench	-	-	-	- / -			0.06 / 0.4

Seq	Operation	How (beginner)	Pieces	Machine	Needle	Thread	SPI	Stitch / seam	mm	Ref	h / SAM min
7.25	Fit the hardware	IM1-IM3 (SK-11, DECISION IM-CAM-TAB): the cam is SEWN to the IMC tab (cam end box), the IMC's other loop carries the G-hook (IM1 / IM2) or the wiregate carabiner (IM3); the IM strap's free end goes through the cam from the wearer side, its sewn loop girth-hitched on the left eye; US1/US2: cam, then the STAINLESS 25 mm tri-glide threaded behind the cam as the cradle slip backstop (orchestrator 2026-11-13; gate US-BS); T3 loops: wiregate carabiner + TPU sleeve; W1-W3: rated G-hook end, heavy ladder lock on the gusset side; W3 is the OUTER layer (BL-043 DECISION W3-OVER): it may lie over the RT straps and the roll, so slacken W3 (G-hook off) before opening the roll top; guys: toggle in the B-25 tab, through the B-27 port, heavy ladder lock on LLT, nylon tri-glide tail keeper; B-28: toggle rear, ladder lock both ends, 2 elastic keepers; RT: thread the free end through the ADJUSTABLE (male) half of each side-release buckle (its female half is already on the RTM gusset tab), K-RT keeper below it; E5: cord lock; cords MK-cord per their use.	hardware	bench	-	-	-	- / -		SK-11; QC-11	1.00 / 7.2
8.01	Scope pod A1	Box pleats in A1-b, leading-edge bar A1-d on the sleeve mouth (EFa), sleeve face to back panel A1-a round the perimeter; 4 hooks / loops to the HPL hard points.	A1-a, A1-b, A1-d	WF1	Nm 110/18 DPx17, round point R	BN tex90	8	301 / SSa2 + EFa	3770		1.63 / 11.8
8.02	Meat liner MS (SWAP slab)	Tube MS-a to frame face MS-b and floor MS-c; HDPE MS-s in the roll pocket. Silpoly: clips, no pins; 10 SPI tex 70.	MS-a, MS-b, MS-c, MS-s x2	WF1	Nm 90/14 DPx17, round point R	BN tex70	10	301 / SSa2	5280		2.08 / 15.0
9.01 V	F-01 fly: panel seams	Join the fly panels (F-01a spine R | F-01b side 1271 mm; F-01a spine L | F-01b side 1271 mm). Silpoly: weights and clips, no pins; 301 then flat-fell; seam-seal the felled seams (piece note: silpoly: seams 301 then flat-felled + seam-sealed; 20 mm elastic channel hems top + bottom)	F-01a, F-01b	WF1	Nm 90/14 DPx17, round point R	BN tex70	10	301 / SSa then LSc flat-fell (ISO 4916 class 2)	7628		3.05 / 22.0
9.02 V	F-01 fly: elastic channel hems	Turn the hems 20 mm on the drawn hem lines (3249 mm of hem). Cut one length of 6 mm elastic (F-EL) per channel: F-01a bottom hem 433 mm: elastic 325 mm; F-01a top hem 385 mm: elastic 289 mm; F-01b bottom hem 201 mm: elastic 151 mm (x2, one per side); F-01b frame hem 1014 mm: elastic 761 mm (x2, one per side) (total 2436 mm). Thread each with a safety pin, bar-tack both ends of the elastic inside its channel (it is NOT lapped into a loop).	F-01a, F-01b, F-EL	WF1	Nm 90/14 DPx17, round point R	BN tex70	10	301 / EFb channel	3248		2.35 / 16.9
10.01	Trim, seal, turn	Trim thread ends to 3 mm, heat-seal webbing ends that frayed, turn the bag right side out through the door.		bench	-	-	-	- / -			0.50 / 3.6
10.02	Final inspection and weigh	QC sheet rows QC-12 to QC-20 (qc_sheet.md); weigh each kit against the PRIMARY mass targets in QC-14.		bench	-	-	-	- / -		QC-12 to QC-20	0.50 / 3.6
10.03	Pack	README.md, Packaging section.		bench	-	-	-	- / -			0.20 / 1.4

7. Load points: bartack and box-X specification

Box sizes are the box-X marks drawn on the pattern pieces. Load = factored design load from the record named (x 3.0 on the 1 g simulation per HA-LS-1 rev C). Capacity by attach_system rules. Proof = the test that releases the point (register).

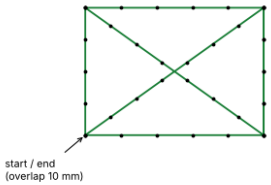
Point	Part	Where	Pattern	Stitches	Machine	Thread / needle	Load N (lbf)	Source	Capacity N	Governs	SF	Proof
port guy, toggle end	TG	TG toggle loop (B-25 tab side)	box-X 21 x 25 mm (0.83 x 0.98 in), path 157 mm	48	BT LK-1900BN	bonded_nylon_tex135 / Nm 140/22 DPx17, round point R	288.0 (65.0)	corrie_f_body.md 13.5: 96.1 N x 3.0	3186.0	box-X	11.06	TL-1 497 N (toggle in the B-25 tab, BL-041)
port guy, bag end	LLT + B-22 + B-02h	inside the hinge band at x +/-130.3	box-X 21 x 25 mm (0.83 x 0.98 in), path 157 mm	48	BT LK-1900BN	bonded_nylon_tex135 / Nm 140/22 DPx17, round point R	288.0 (65.0)	corrie_f_body.md 13.5: 96.1 N x 3.0	1024.0	B-22 perimeter 50 mm (primary body)	3.56	LL-H 432 N; PT-5 (lock + keeper); alternate body fabric: B-22 seam 1408 N
port guy, B-27 slit ends	B-01	both ends of each slit, z 595 / 745	bartack 10 x 2 mm, 304	28	BT LK-1900BN	bonded_nylon_tex90 / Nm 110/18 DPx17, round point R	- (-)	no rated load (lap only); port mock-up: 100 N guy pull through the lap		-		port mock-up (13.9)
crown strap tab (B-28s x2)	B-28s	toggle loop	box-X 21 x 15 mm (0.83 x 0.59 in), path 124 mm	39	BT LK-1900BN	bonded_nylon_tex135 / Nm 140/22 DPx17, round point R	288.0 (65.0)	corrie_f_body.md 15.2: 288 N envelope (port guy) if a crown strap ever holds the top	2504.0	box-X	8.69	LL-H 432 N (1.5 x 288 N; BL-031 dropped the 20 mm gate)
crown strap tab (B-28t x4)	B-28t	lock tab	box-X 21 x 20 mm (0.83 x 0.79 in), path 140 mm	43	BT LK-1900BN	bonded_nylon_tex135 / Nm 140/22 DPx17, round point R	288.0 (65.0)	corrie_f_body.md 15.2: 288 N envelope (port guy) if a crown strap ever holds the top	2836.0	box-X	9.85	LL-H 432 N (1.5 x 288 N; BL-031 dropped the 20 mm gate)
crown strap front tab on the band	B-02h	B-28 front lock tab, OUTSIDE the hinge band (DECISION B28-FACE, BL-034), box through tab + band	box-X 21 x 20 mm (0.83 x 0.79 in), path 140 mm	43	BT LK-1900BN	bonded_nylon_tex135 / Nm 140/22 DPx17, round point R	288.0 (65.0)	corrie_f_body.md 15.2 (as above)	2836.0	box-X	9.85	LL-H 432 N
B-25 bridle toggle tab	B-25	toggle tab R: tip-end box, clear loop for the POM toggle (DECISION B25-BOX, BL-034); 38 mm bridle, sleeve x 26.5-64.5 (BL-041)	box-X 21 x 15 mm (0.83 x 0.59 in), path 124 mm	39	BT LK-1900BN	bonded_nylon_tex135 / Nm 140/22 DPx17, round point R	323.0 (73.0)	crown_tube_guy_loop.md 2: guy + B-28 crown strap on one tab, envelope 323 N	439	crown bar wall under the 38 mm sleeve (basis A, knocked down)	1.36	TL-1 497 N on real bar stock (crown_tube_guy_loop.md 6)
B-25 bridle toggle tab	B-25	toggle tab L: tip-end box, clear loop for the POM toggle (DECISION B25-BOX, BL-034); 38 mm bridle, sleeve x 26.5-64.5 (BL-041)	box-X 21 x 15 mm (0.83 x 0.59 in), path 124 mm	39	BT LK-1900BN	bonded_nylon_tex135 / Nm 140/22 DPx17, round point R	323.0 (73.0)	crown_tube_guy_loop.md 2: guy + B-28 crown strap on one tab, envelope 323 N	439	crown bar wall under the 38 mm sleeve (basis A, knocked down)	1.36	TL-1 497 N on real bar stock (crown_tube_guy_loop.md 6)
B-25 bar sleeve laps	B-25	bar sleeve R lap (bartack); leaves the bar DOWN, aft side; bar sleeve L lap (bartack) (sewn at frame assembly, round the crown top bar)	bartack 16 x 2 mm, 304	42	BT LK-1900BN	bonded_nylon_tex90 / Nm 110/18 DPx17, round point R	- (-)	no rated load: the sleeve wraps the bar, the lap only keeps it closed		-		LL-H toggle-loop pull (whole bridle)
IM3 bag-hook tongue	T3	loop 34x30	box-X 34 x 30 mm (1.34 x 1.18 in), path 219 mm	65	BT LK-1900BN	bonded_nylon_tex135 / Nm 140/22 DPx17, round point R	1702.0 (383.0)	loadpath_verification A1: IM3 1702 N factored (guys-off misuse)	4000.0	carabiner (PT-1 basis: 0.5 x 8 kN minor axis)	2.35	PT-1 2553 N carabiner; PT-2 2553 N
IM3 bag-hook tongue	T3	box-X to B-17 34x34	box-X 34 x 34 mm (1.34 x 1.34 in), path 232 mm	68	PS AMS-210EN (or WF2 + J1)	bonded_nylon_tex135 / Nm 140/22 DPx17, round point R	1702.0 (383.0)	loadpath_verification A1: IM3 1702 N factored (guys-off misuse)	4000.0	carabiner (PT-1 basis: 0.5 x 8 kN minor axis)	2.35	PT-1 2553 N carabiner; PT-2 2553 N

Point	Part	Where	Pattern	Stitches	Machine	Thread / needle	Load N (lbf)	Source	Capacity N	Governs	SF	Proof
B-17 IM3 tab perimeter (double row)	B-17	OUTSIDE B-01 at x +/-115, z 385-455	double-row perimeter 3 / 9 mm, 301		WF2	bonded_nylon_tex90 / Nm 110/18 DPx17, round point R	1702.0 (383.0)	loadpath_verification A1	3277.0	perimeter 80 mm loaded, 2 rows	1.93	PT-2 2553 N on the 70 mm tab; perimeter 3 / 9 mm, 8 SPI tex 90; capacity = patch_seam on the 80 mm loaded top edge (the same for the 70 or 110 mm tall tab: the side rows are not credited); SF PENDING PT-2 on the 70 mm tab (BL-040 m-3)
IM fairlead eyes (6)	IME	frame side, captive in the printed fairleads	box-X 34 x 30 mm (1.34 x 1.18 in), path 219 mm	65	BT LK-1900BN	bonded_nylon_tex135 / Nm 140/22 DPx17, round point R	1702.0 (383.0)	IM3 eye carries the bag hook (A1 1702 N); IM1/IM2 605 N (S1)	4430.0	box-X	2.6	MT-1 analogue at 38 mm; IM3-EP 2553 N
hinge eye	HE + B-18 + B-01	x +/-115, box 4-11 mm above the floor seam line; B-18 and both HE raw ends in the B-01 | B-05 seam (DECISION HE-ONE)	box-X 34 x 7 mm (1.34 x 0.28 in), path 151 mm	46	BT LK-1900BN	bonded_nylon_tex135 / Nm 140/22 DPx17, round point R	1019.0 (229.0)	load_standard.md (rev C) bag-on hinge case 1019 N factored; bag_attachment.md section 5 (loadpath A5 795 N is the sim-only value)	2355.0	B-18 seam (80 mm, B-05 strip)	2.31	PT-3 1530 N (= 1.5 x 1019) on the printed fairlead; SEW-F HE 625 N x 1e5; box-X 34 x 7 capacity 3067 N (attach_system.box_x thread path); the box top row PEELS (BL-044 he_peel): leg B leaves it 23 deg off the box line in line, 50 deg with B-01 held at the cheek tips = 0.20-0.38 x the hinge load across the row; no peel capacity is credited: rated by PT-3 on the printed fairlead (worst peel) and SEW-F HE
IM1 end loop	IM1	left end loop, girth-hitched on the eye (box drawn at the tip end)	box-X 21 x 25 mm (0.83 x 0.98 in), path 157 mm	48	BT LK-1900BN	bonded_nylon_tex135 / Nm 140/22 DPx17, round point R	605.0 (136.0)	loadpath_verification S1: 605 N	3186.0	box-X	5.27	MT-1 908 N; SEW-F (box-X fatigue); MT-1 specimens are sewn exactly this way
IM2 end loop	IM2	left end loop, girth-hitched on the eye (box drawn at the tip end)	box-X 21 x 25 mm (0.83 x 0.98 in), path 157 mm	48	BT LK-1900BN	bonded_nylon_tex135 / Nm 140/22 DPx17, round point R	605.0 (136.0)	loadpath_verification S1: 605 N	3186.0	box-X	5.27	MT-1 908 N; SEW-F (box-X fatigue); MT-1 specimens are sewn exactly this way
IM3 end loop	IM3	left end loop, girth-hitched on the eye (box drawn at the tip end)	box-X 21 x 25 mm (0.83 x 0.98 in), path 157 mm	48	BT LK-1900BN	bonded_nylon_tex135 / Nm 140/22 DPx17, round point R	605.0 (136.0)	loadpath_verification S1: 605 N	3186.0	box-X	5.27	MT-1 908 N; SEW-F (box-X fatigue); MT-1 specimens are sewn exactly this way
IM cam anchor tab	IMC	cam end (MT-1) 21x25: dog-bone between the cam bar and the hook / carabiner (SK-11)	box-X 21 x 25 mm (0.83 x 0.98 in), path 157 mm	48	BT LK-1900BN	bonded_nylon_tex135 / Nm 140/22 DPx17, round point R	605.0 (136.0)	loadpath_verification S1: 605 N (same strap)	3186.0	box-X	5.27	MT-1 908 N (2 IMC specimens added to the MT-1 lot)
IM cam anchor tab	IMC	hook end (MT-1) 21x25: dog-bone between the cam bar and the hook / carabiner (SK-11)	box-X 21 x 25 mm (0.83 x 0.98 in), path 157 mm	48	BT LK-1900BN	bonded_nylon_tex135 / Nm 140/22 DPx17, round point R	605.0 (136.0)	loadpath_verification S1: 605 N (same strap)	3186.0	box-X	5.27	MT-1 908 N (2 IMC specimens added to the MT-1 lot)
US cam anchor	USA	pin end (MT-1) 21x25: dog-bone between the right pin and the cam bar	box-X 21 x 25 mm (0.83 x 0.98 in), path 157 mm	48	BT LK-1900BN	bonded_nylon_tex135 / Nm 140/22 DPx17, round point R	720.0 (162.0)	loadpath_verification S5-S7: 720 N cradle (the same strap as the left eye)	3186.0	box-X	4.43	MT-1 1080 N (2 USA specimens added to the MT-1 lot)
US cam anchor	USA	cam end (MT-1) 21x25: dog-bone between the right pin and the cam bar	box-X 21 x 25 mm (0.83 x 0.98 in), path 157 mm	48	BT LK-1900BN	bonded_nylon_tex135 / Nm 140/22 DPx17, round point R	720.0 (162.0)	loadpath_verification S5-S7: 720 N cradle (the same strap as the left eye)	3186.0	box-X	4.43	MT-1 1080 N (2 USA specimens added to the MT-1 lot)

Point	Part	Where	Pattern	Stitches	Machine	Thread / needle	Load N (lbf)	Source	Capacity N	Governs	SF	Proof
US1 sewn eye	US	eye on the left pin (US-EYE fold, pattern)	box-X 21 x 25 mm (0.83 x 0.98 in), path 157 mm	48	BT LK-1900BN	bonded_nylon_tex135 / Nm 140/22 DPx17, round point R	720.0 (162.0)	loadpath_verification S5-S7: 720 N; BL-050 B lane span: 529 N at the 53 kg float limit, us_route.json cradle_span	3186.0	box-X	4.43	MT-1 1080 N; SEW-F (box-X fatigue); US-BS; ; cam side: stainless tri-glide backstop behind the cam (orchestrator 2026-11-13; gate US-BS)
US2 sewn eye	US2	eye on the left pin (US-EYE fold, pattern)	box-X 21 x 25 mm (0.83 x 0.98 in), path 157 mm	48	BT LK-1900BN	bonded_nylon_tex135 / Nm 140/22 DPx17, round point R	720.0 (162.0)	loadpath_verification S5-S7: 720 N; BL-050 B lane span: 639 N at the 53 kg float limit, us_route.json cradle_span	3186.0	box-X	4.43	MT-1 1080 N; SEW-F (box-X fatigue); US-BS; ; cam side: stainless tri-glide backstop behind the cam (orchestrator 2026-11-13; gate US-BS)
W1 G-hook end	W1	G-hook fold 2 x 30 mm	box-X 21 x 25 mm (0.83 x 0.98 in), path 157 mm	48	BT LK-1900BN	bonded_nylon_tex135 / Nm 140/22 DPx17, round point R	225.0 (51.0)	loadpath_verification S9: 225 N	3186.0	box-X	14.16	LL-H 432 N; W-PULL 338 N
W2 G-hook end	W2	G-hook fold 2 x 30 mm	box-X 21 x 25 mm (0.83 x 0.98 in), path 157 mm	48	BT LK-1900BN	bonded_nylon_tex135 / Nm 140/22 DPx17, round point R	225.0 (51.0)	loadpath_verification S9: 225 N	3186.0	box-X	14.16	LL-H 432 N; W-PULL 338 N
W3 G-hook end	W3	G-hook fold 2 x 30 mm	box-X 21 x 25 mm (0.83 x 0.98 in), path 157 mm	48	BT LK-1900BN	bonded_nylon_tex135 / Nm 140/22 DPx17, round point R	225.0 (51.0)	loadpath_verification S9: 225 N	3186.0	box-X	14.16	LL-H 432 N; W-PULL 338 N
W haul extender	WX	box-X hook end	box-X 21 x 25 mm (0.83 x 0.98 in), path 157 mm	48	BT LK-1900BN	bonded_nylon_tex135 / Nm 140/22 DPx17, round point R	225.0 (51.0)	ASSUMPTION WX-LOAD: same float-clamp load as S9 (225 N)	3186.0	box-X	14.16	LL-H; one extender pulled to 338 N
W haul extender	WX	box-X loop	box-X 21 x 25 mm (0.83 x 0.98 in), path 157 mm	48	BT LK-1900BN	bonded_nylon_tex135 / Nm 140/22 DPx17, round point R	225.0 (51.0)	ASSUMPTION WX-LOAD: same float-clamp load as S9 (225 N)	3186.0	box-X	14.16	LL-H; one extender pulled to 338 N
B-24 float link: hook end	B-24	hook-end fold (boxes as drawn by attach_system.link_piece)	box-X 34 x 30 mm (1.34 x 1.18 in), path 219 mm	65	BT LK-1900BN	bonded_nylon_tex135 / Nm 140/22 DPx17, round point R	941.0 (212.0)	loadpath_verification S12 / RD-02: 941 N misuse	4000.0	carabiner	4.25	PT-6 1410 N
B-24 float link: loop bases	B-24	3 loop bases (boxes as drawn by attach_system.link_piece)	box-X 34 x 20 mm (1.34 x 0.79 in), path 187 mm	56	BT LK-1900BN	bonded_nylon_tex135 / Nm 140/22 DPx17, round point R	941.0 (212.0)	loadpath_verification S12 / RD-02: 941 N misuse	3786.0	box-X	4.02	PT-6 1410 N; loop boxes close each T3 loop across both plies
roll-top clip (RT)	RT	strap end box-X'd on the hem B-07, free end at the clip-path notch (BL-041 RT-FWD)	box-X 21 x 25 mm (0.83 x 0.98 in), path 157 mm	48	BT LK-1900BN	bonded_nylon_tex135 / Nm 140/22 DPx17, round point R	- (-)	UNRATED: hand cinch on the roll-top closure, no load case (not on an HA-LS-1 path)	3186.0	box-X		QC-11 hand pull; buckle record 450 N (ESTIMATE) governs anything put on it
roll-top clip (RTM)	RTM	female buckle tab box-X'd on the gusset B-03, 60 mm aft of the frame seam, 202 mm under the collar seam (BL-041 RT-FWD)	box-X 21 x 25 mm (0.83 x 0.98 in), path 157 mm	48	BT LK-1900BN	bonded_nylon_tex135 / Nm 140/22 DPx17, round point R	- (-)	UNRATED: hand cinch on the roll-top closure, no load case (not on an HA-LS-1 path)	3186.0	box-X		QC-11 hand pull; buckle record 450 N (ESTIMATE) governs anything put on it

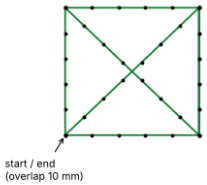
SK-1 Load-point stitch patterns (true proportions; dots = 7 SPI stitch positions)

box-X 21 × 15 mm (0.83 × 0.59 in)



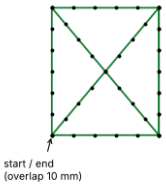
39 stitches incl. 4 lock, 7 SPI, tex 135, Nm 140 R
used on: B-25, B-28s
* B-25: see the note at the foot

box-X 21 × 20 mm (0.83 × 0.79 in)



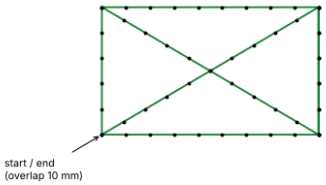
43 stitches incl. 4 lock, 7 SPI, tex 135, Nm 140 R
used on: B-02h, B-28t

box-X 21 × 25 mm (0.83 × 0.98 in)



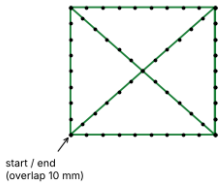
48 stitches incl. 4 lock, 7 SPI, tex 135, Nm 140 R
used on: IM1, IM2, IM3, IMC, LLT + B-22 + B-02h, RT, RTM, TG, US, US2, USA, W1, W2, W3, WX

box-X 34 × 20 mm (1.34 × 0.79 in)



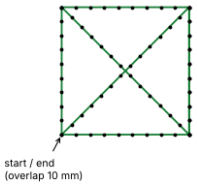
56 stitches incl. 4 lock, 7 SPI, tex 135, Nm 140 R
used on: B-24

box-X 34 × 30 mm (1.34 × 1.18 in)



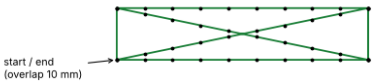
65 stitches incl. 4 lock, 7 SPI, tex 135, Nm 140 R
used on: B-24, IME, T3

box-X 34 × 34 mm (1.34 × 1.34 in)



68 stitches incl. 4 lock, 7 SPI, tex 135, Nm 140 R
used on: T3

box-X 34 × 7 mm (1.34 × 0.28 in)



46 stitches incl. 4 lock, 7 SPI, tex 135, Nm 140 R
used on: HE + B-18 + B-01

BT-S bartack 10 × 2 mm, 28 st (304)



LK-1900 standard tack, tex 90, Nm 110 R (unrated)
B-27 slit ends, FP mouth corners (BL-043: W3G deleted; every BT-S is 10 × 2)
no zigzag: BT-301 = 4 straight passes, 2 mm stitch

BT-M bartack 16 × 2 mm, 42 st (304)

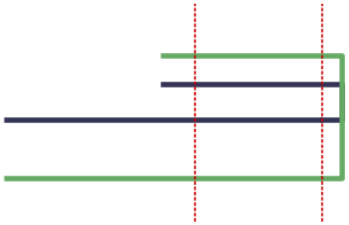


LK-1900 standard tack, tex 90, Nm 110 R (unrated)
I2K, HE neck (1 per leg), IME neck (op 2.05), US neck (op 7.15), USA neck (op 7.16), lid channel ends, B-25 sleeve laps, B-25 neck fold and mitre (BL-048)
no zigzag: BT-301 = 4 straight passes, 2 mm stitch

* box-X 21 × 15: EXCEPTION (B25-BOX): B-25 is 38 mm webbing; this 21 mm box leaves 8.5 mm unsewn each edge so the 13 mm toggle loop stays open

SK-2 Seam constructions (cross-sections, not to scale; red dashed = stitch row)

SSa2 body seam: 12 mm, then allowances turned and caught at 6 mm (2 rows)



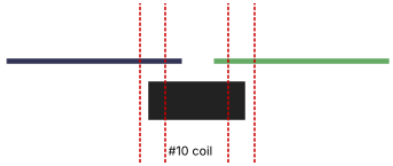
LSd Hypalon patch: double row 3 / 9 mm from the edge, 8 SPI tex 90



EFb roll-top hem round the HDPE strip



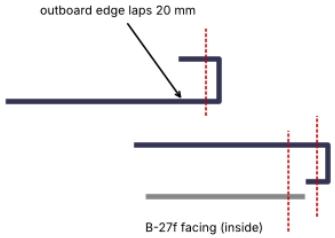
ZIP: tape sewn at 5 mm, turned, edge-stitched 3 mm (door, lid, pockets)



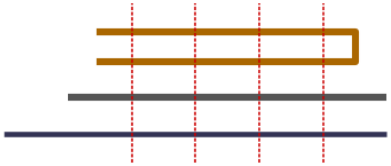
SSa-ins: loop / tab caught in the seam, backtack each side (HPL, K-W, E5)



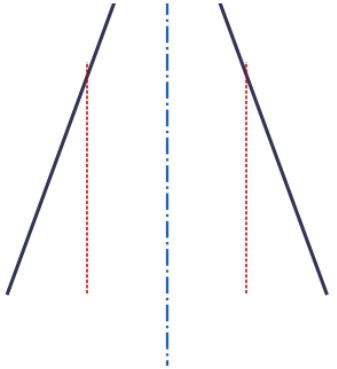
B-27 guy port: both edges turned 6 mm, facing on the inboard edge, bartack ends



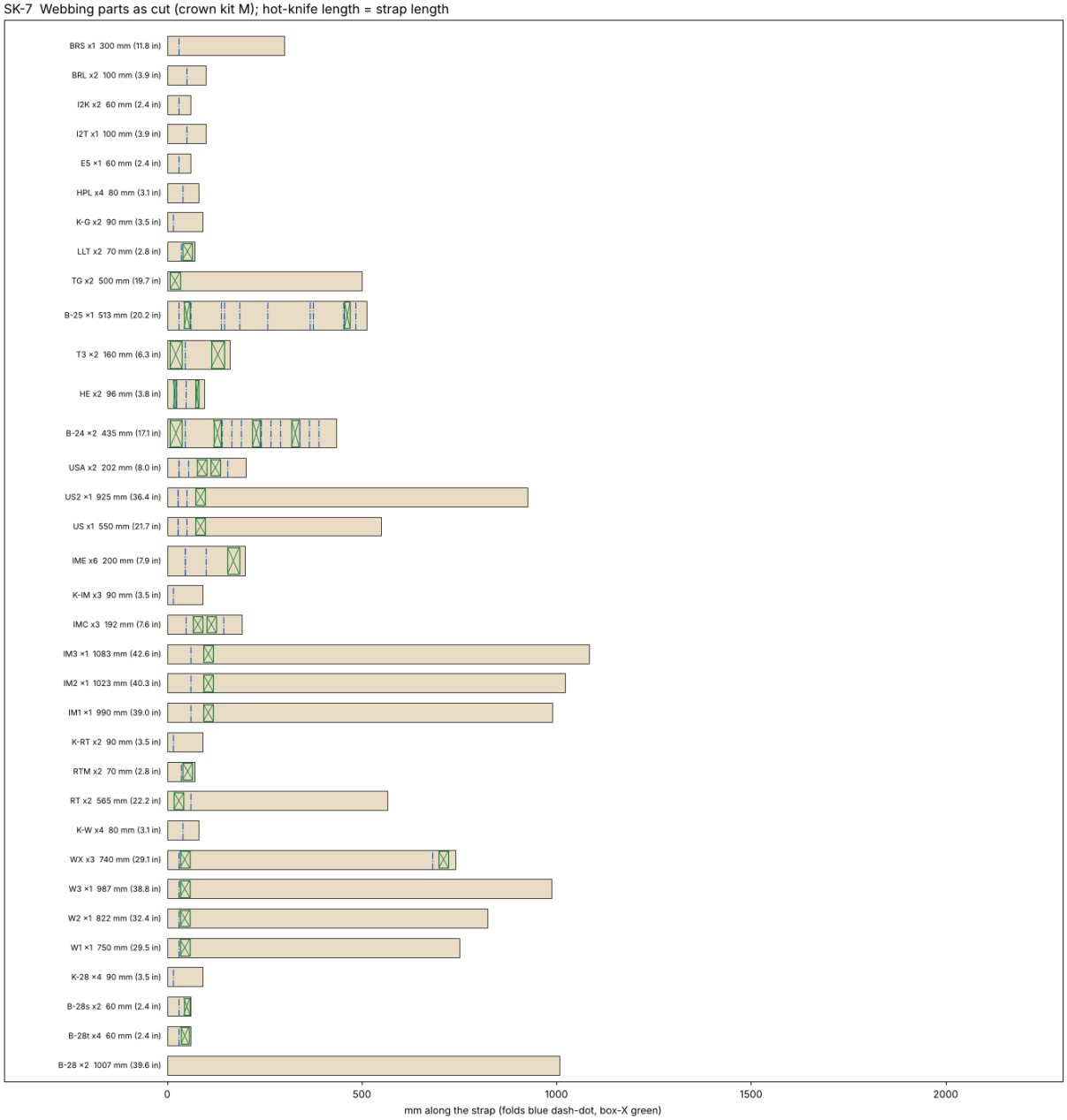
Load stack (e.g. LLT + B-22 + B-02h): webbing fold / Hypalon / panel, box-X through all



DART (B-29a, 4 corners): fold on the centre, sew leg to leg, tie off at the apex



SK-7 Webbing parts as cut (crown kit M); hot-knife length = strap length



8. QC checkpoints, release tests, defects

ID	Stage	Check	Method	Pass	Frequency
QC-01	Print	Test square	ruler on 00_PRINT_FIRST and on one page per print run	100.0 +/- 1.0 mm both ways	every print run
QC-02	Incoming	Fabric weight and width	cut 100 x 100 mm, weigh 0.01 g; tape the roll width	gsm within +/- 5 % of the record; width >= usable_width_mm	every roll
QC-03	Cutting	Piece size and cut-outs	lay the pattern on the cut piece	+/- 1 mm on pieces < 300 mm, +/- 2 mm on larger; windows / slots / slits present and clean	first piece of each lay + 1 in 10
QC-04	Cutting	Notches, grain, webbing length	visual; tape	notch 6 mm (never past the seam line); grain within 2 deg of the arrow; webbing +/- 2 mm, ends sealed	first of each + 1 in 10
QC-05	Sewing	Stitch density	count stitches over 25.4 mm	body 8, light 10, box-X 7 SPI; +/- 1	start of each shift / each op set-up
QC-06	Sewing	Tension balance	cut a swatch seam open	lock mid-stack, no loops on either face	each set-up
QC-07	Load points	Box-X and bartacks	visual against SK-1; count	full rectangle + both diagonals, start / end overlap >= 10 mm, stitch count >= load table, no skipped stitch, needle hole not torn	100 %
QC-08	Assembly	Seam match	notches at both ends of every 50 mm	notch offset <= 2 mm; seam end mismatch <= 3 mm; allowance 12 +/- 1 mm	100 %
QC-09	Assembly	Door zip	run both sliders 10 x end to end; look at the corners	no snag, no wave > 3 mm at the R 55 corners, tape not caught, stops at the hinge band	100 %
QC-10	Assembly	B-27 ports	pull the guy through at 45 deg; look inside	20 mm lap closes flat; both ends bartacked; facing follows the crease at z 631	100 %
QC-11	Hardware	Hardware fitted	count against the BOM; threading as SK-7	every lock threaded the right way (pull test by hand), carabiners gate-in, cord lock holds	100 %
QC-12	Final	Width	tape across the widest point, bag packed full	<= 394 mm (15.5 in) (owner limit)	first article + 1 in 10
QC-13	Final	Volume	fill with 10 mm foam balls / litre jug method, collar rolled 3 times	owner limit 85 L +/- 3 L (82-88 L, 5200 cu in). EXPECTED on the sewn bag: 81.9-84.0 L (fine-mesh closed volume, corrie_f_body.md 13.5), so a reading of 81.9-82.0 L is the design, not a sewing fault: record it and tell the pack-designer (BL-040 m-17)	first article
QC-14	Final	Mass by kit, PRIMARY fabrics (the set you build)	1 g scale, bag empty, kit by kit	base 1226 g (43.2 oz), as hunted 1530 g, meat trip 1735 g, each +/- 3 % (= ledger on the team B basis 1161 / 1443 / 1639 g + the primary pattern delta of each kit, costing.md). Alternate-fabric bags: the ledger values	every bag
QC-15	Final	No hook-and-loop, coil zips only	visual	zero hook-and-loop anywhere (owner: quiet bag)	100 %
QC-16	Final	Camo use	visual	camo only on B-02u, SH-07, SH-01b, FP-b, B-29f, A1-d (accents)	100 %
QC-17	Final	Thread ends, seals	visual	ends <= 3 mm, no open webbing ends, no burnt fabric	100 %
QC-18	Final	Labels	visual	ID label with rev 3 + date, size / crown kit, load-limit card (53 kg frame meat, RD-02)	100 %
QC-19	Final	Kits complete	count	base / as-hunted (lid, B-28 per kit, scope pod) / meat-trip (WX, B-24, liner) bags each complete	100 %
QC-20	Final	Fit on the frame	hook to a frame or the fit jig: hinge pins, IM eyes through B-12 ports, guys through B-27	every eye reaches its port without strain; door opens fully; hinge eyes on the z 60 pins with the pin centre 22-29 mm above the floor seam (the HE-FIT band, already proven before cut; this confirms the batch)	first article + 1 in 25

Physical tests that gate release

ID	Test	Gates	Steps	Pass / fail	Source
MT-1	25 mm box-X bartack proof at the BL-010 per-row loads	every 25 mm box-X on IM1-IM3 and US1/US2	sew the loops exactly like the pack (necks folded and tacked first); loop over the 5 mm pin; 5 loops to 908 N (92.6 kg) and 5 to 1080 N (110.1 kg); the 2 IMC tabs (cam bar on a 5 mm pin each end) to 908 N; the 2 USA anchors (necked loop on a 5 mm pin, cam bar in the other) to 1080 N; hold 1 min each	5 of 5 hold 908 N (IM) and 5 of 5 hold 1080 N (US) for 1 min, 2 of 2 IMC hold 908 N, 2 of 2 USA hold 1080 N, no broken stitch	test_register.csv
MT-2	meat-strap hook proof per strap (BL-010)	IM free-end hooks	as PT-1: G-hooks to 645 N (65.7 kg, covers IM1 410 N), carabiners to 910 N (92.8 kg); hold 1 min	5 of 5 G-hooks hold 645 N and 5 of 5 carabiners hold 910 N for 1 min, gates close	test_register.csv
MT-3	25 mm cam buckle slip at the BL-010 per-row loads, dry, wet and cold	IM / US cams	thread the cam as in the pack and paint-pen mark at the cam mouth; DRY: pull to 908 N (92.6 kg), hold 1 min, measure; then to 1080 N (110.1 kg), hold 1 min, measure. WET: fresh webbing, cam + webbing soaked 10 min, same two holds. COLD: fresh webbing, 2 h in the freezer, write the freezer reading, pull within 2 min of taking it out, same two holds. A home freezer at -18 C or colder is ACCEPTED as the record's -20 C (ASSUMPTION FREEZER)	mark moves < 3 mm at 908 N and at 1080 N, 5 of 5, in each of dry / wet / cold	test_register.csv
PT-1	IM3 bag carabiner on the sewn T3 tongue, per-row proof (BL-010)	IM3 carabiner on T3	carabiner over the 5 mm steel pin; sewn T3 tongue on the carabiner; pull with the come-along to 2553 N (260.2 kg on the scale); hold 60 s; release; check the gate closes and look for set	5 of 5 hold 2553 N for 60 s, gate closes after, no visible set	test_register.csv
PT-2	T3 tongue on the 70 mm B-17 tab on a B-01 swatch (BL-010 load)	T3 on the 70 mm B-17 tab	clamp the B-01 swatch 120 mm wide between bar clamps; tongue loop on the pin; pull to 2553 N (260.2 kg); hold 60 s	5 of 5 hold 2553 N, no stitch break, no laminate tear	test_register.csv
PT-3	HE hinge eye on B-18 + B-01 + B-05 floor seam, sewn as op 4.05 (HE-ONE / HE-PIN), no binding	HE eye on B-18 (no binding)	FIXTURE: a fresh print of `print/frame_h1/test_pieces/hefit_hinge_fairlead_R.stl` (not the HE-FIT piece), eye reamed to 7.2, its UHMWPE sleeve, bolted upright to the bench plate through the stay segment; flat spacer plate (PLA, or 6 mm ply shimmed) screwed to the printed stay segment, its face flush with the cheek tips (17.7 mm proud of the stay surface) from the cheeks down past the box, so B-01 is HELD at the cheek-tip plane and the box cannot swing in line: leg B leaves the box top at the worst peel angle (he_peel 'tip_plane'). The pin centre is then 7.6 mm toward the frame from the B-01 face and 14 mm above the box top (the bag geometry, DECISION HE-ENV). Put the the hardened steel M5 x 40 dowel (DIN 6325 m6, HRC 58-62; NOT a 7075 frame pin) through cheek, sleeve, eye, cheek (only the 18 mm neck between the cheeks); clamp the B-05 strip ALONE in the bar clamp below; the B-01 top edge stays FREE (never clamped: a clamp there carries the load past the floor seam); pull DOWN along the spacer face (leg B leaves the box top at about 50 deg: the peel the frame gives) to 1530 N (156.0 kg); hold 60 s; with the loupe check the box TOP row first, then the neck tacks and the floor seam. PIN CHECK (BL-048 M-C): roll the dowel on the bench plate before and after the test: it rolls freely with no visible wobble and a 0.05 mm feeler does not pass under its middle; a bent pin voids the specimen. If the printed cheeks crack before the eye fails, the specimen is void: re-run it in a machined aluminium copy of the same section (ASSUMPTION PT3-PRINT). PRINT (BL-048 m-6): fixtures print as the H1 frame: PAHT-CF, dried, the H1 slicer settings (print/frame_h1/README.md), annealed as H1.	5 of 5 hold 1530 N 60 s: no stitch break (box top row included), no Hypalon or laminate tear, B-18 still in the floor seam, the eye still open (no row across it)	test_register.csv

ID	Test	Gates	Steps	Pass / fail	Source
HE-FIT	hinge eye 1:1 dry fit on the printed BL-042 z 60 hinge fairlead (eye on the pin, neck in the throat)	hinge eye on the z 60 pin of the printed BL-042 hinge fairlead, 1:1 dry fit before PT-3 and before cut (BL-041 X-2, BL-044 DECISION HE-ENV); step 0 = gate US-ROUTE on the US test piece (BL-048 N-1)	(0) US ROUTE (BL-048 N-1, re-run on the BL-050 lanes; pass criterion below): print the US test piece `designs/pack_5200/print/frame_h1/test_pieces/us_route_stay_R.stl` (R stay z 40-160 with the hinge fairlead and the US1 / US2 lane lugs, cut from `h1_assembled.stl` by `us_route.export_test_piece`; BL-050: printed WITHOUT the US2 outer cheek) and the bolt-on clevis `designs/pack_5200/print/frame_h1/us2_clevis_R.stl`; print both as the H1 frame: PAHT-CF, dried, H1 settings, annealed; ream the clevis bolt holes 5.1 and bolt the clevis on with 2 Ti M5 x 40 DIN 7984, 2 A2 nuts in the side traps and 4 x 10 mm OD DIN 433 washers (NOT the z 234 joint's M5 x 45 / 12 mm washers; shopping list), blue threadlocker, snug; clamp the piece upright; sew two USA anchors (25 mm, as op 7.16: the piece is the R stay, where the cam anchors sit) and hang the HE specimen of this test on its hinge pin (the hardened steel M5 x 40 dowel (DIN 6325 m6, HRC 58-62; NOT a 7075 frame pin) in each fairlead); USA on the US1 pin, its cam on a 25 mm strap laid down the B-01 offcut over the piece, round a 20 mm dowel at the floor-seam line (the floor corner, 25 mm under the hinge pin) to the crane scale, cinched to 444 N (45.3 kg); then the same from the US2 pin at 296 N (30.2 kg), the dowel at the strap's own lane (straight under its pin: the B-16 / B-16b channel ends are at the lane x, BL-050 B); repeat with the IME eye on the IM3-EC piece pulled at -82 to -88 deg (the sim directions); then slide the US2 eye by hand across the clevis seam in the throat floor under 5 kg (SEAM-FLOOR: it must not catch; chamfer 0.3 mm both edges if it does). PASS (gate US-ROUTE): at the cinched load, along its whole run from the eye to the dowel, no part of the strap, eye or box touches the hinge fairlead, the HE (eye, legs or box) or the other US fairlead: a 10 mm feeler block passes between them everywhere (the design clearance is >= 10 mm). FAIL = any contact or a gap under 10 mm. On the BL-050 print with the BL-050 B channel ends the model gives worst clearances US1 12.4 / US2 14.6 mm over the 15 / 19 / 23 in meat drapes, both sides (us_route.json 'now'; the BL-048 channel drawing at x +/-95 rode the hinge cheeks, bearing about 72 N); the bag is not cut until it passes; (1) re-caliper ONE ply of the 38 mm roll in hand at 5 places (the first caliper is at the PURCHASE, on a sample, shopping list; BL-048 M-B): every reading <= 1.6 mm, else reject the roll (3 plies of a thicker web leave < 0.7 mm under the pin); (2) clamp the printed stay segment upright; (3) slide the neck round the pin by hand: the pin goes through cheek, sleeve, eye, cheek with the eye seated, no forcing; then push the neck hard against each cheek in turn and slide it round the pin again (its folded edge passes the printed throat-corner fillet: 0.3 mm clear on the model, BL-044 he_stl_check); (4) hang 5 kg from the B-05 strip so the eye loads the pin; (5) caliper from the floor seam line to the pin centre; (6) feeler gauge between the box top and the cheeks' lower edge; (7) check the eye is not folded over the box and the neck is not pinched by the cheeks; (8) BL-048 DECISION HE-NECK-GAUGE: 5 sewn necks (HE, IME, US) each pass the 18.6 mm go / no-go slot, and the printed throat admits a 19.6 mm gauge block (then the widest neck keeps >= 0.5 mm a side); photo	ply <= 1.6 mm; pin centre 22-29 mm above the floor seam line; >= 2 mm feeler between the box top and the cheeks; the eye slides round the pin by hand; neck alone in the 20 mm throat; eye points up (no fold-down)	test_register.csv

ID	Test	Gates	Steps	Pass / fail	Source
TL-1	guy + crown-strap toggles in the 38 mm B-25 sleeve on the real crown bar, 497 N	B-25 38 mm sleeve + guy and crown-strap toggles on the real crown bar at 497 N, two rigs: A guy 22 deg below horizontal, B 58 deg and 13.6 % outboard, crown strap 43 N, run anchored on a spine stub (BL-047 record bl047_frame_items.json tl1; BL-048 M-A)	clamp the tube ends in V-blocks >= 40 mm outside the sleeve edges (the bar reacts as a beam), NO cradle or saddle under the sleeve; anchor the run's spine end on the spine stub 25.0 mm fore of the bar axis under the 6.0 mm collar (BL-046 geometry; the run then pulls fore-down by itself); dial bore gauge 10-18 mm range, 0.01 mm, >= 110 mm reach (BL-043 m-7: basis A predicts dD at the LOOP CENTRE; the gauge goes in through the open outboard tube end, anvils on the pull axis at the sleeve centre, depth marked on the shaft; zero it on the unloaded tube); TWO RIGS (BL-047 record, bl047_frame_items.json tl1.pull): specimens 1-3 on RIG B, 4-5 on RIG A, proof 497 N on the guy toggle (the larger of the BL-046 vector 462 N and BL-045's 497 N): RIG A (the governing BL-046 vector): guy toggle pulled 22 deg below horizontal, 0.5 % outboard (unit: down 0.38, aft 0.93, outboard 0.01; come-along anchor 1 m from the toggle at 377 mm below, 926 mm aft, 5 mm outboard); RIG B (run loaded, BL-045's worst guy): 58 deg below horizontal, 13.6 % outboard (down 0.85, aft 0.51, outboard 0.14; anchor at 849 below, 510 aft, 136 outboard); in BOTH rigs the crown-strap toggle carries 43 N (4.4 kg on a cord over a pulley, up and aft as the crown strap) through the proof; the rig's design guy load is 454 N, proofed to 497 N; check each angle with a phone inclinometer on the line; pull in steps 100 / 200 / 300 / 400 / 497 N (50.7 kg), reading the gauge at each step; hold 497 N 60 s, listening (phone microphone on the tube); unload, remove the sleeve, micrometer the loop centre at 0 and 90 deg; check the mitre and the run did not slip (pen marks across the fold and at the stub); loupe + tap test against the control; repeat 1 sample after 4 h in the freezer (POM)	5 of 5: (1) holds 497 N 60 s, toggle does not pull through or crack, no broken stitch; (2) no crackle; (3) gauge at 497 N within 1.15 x the straight line through the 100-300 N readings, no knee; (4) ovalisation at the LOOP CENTRE (bore gauge, BL-043 m-7) <= 0.40 mm at 497 N (basis A 0.39, B 0.2, C 0.005 mm, all at the loop centre); (5) out-of-round after unloading <= 0.02 mm more than before; (6) no crush mark, whitening or resin crack at 10x, tap test clean; the cold sample holds	test_register.csv
PT-5	guy ladder lock + tri-glide keeper slip	guy lock + tri-glide keeper	mark webbing; 10 cycles 0 -> 170 N -> 0 by hand on the come-along; measure the mark	slip < 5 mm over 10 cycles	test_register.csv
PT-6	B-24 float stand-off link	B-24 float link	link hook on the pin; 38 mm loop through the 220 mm loop; pull in line to 1410 N; hold 60 s	5 of 5 hold 1410 N, no box-X break, gate closes	test_register.csv
LL-H	heavy 25 mm ladder lock slip at 432 N, dry, wet and cold (guys, W1-W3, B-28)	heavy 25 mm ladder lock slip at 432 N (1.5 x 288 N), dry, wet and -20 C: guys, W1-W3 and the B-28 crown straps (the B-28 slip gate since BL-031)	thread as in the pack and paint-pen mark at the lock; DRY at room temperature: hang 44.0 kg (432 N, sand bags) 50 mm above the floor for 1 h, all 5 side by side, then measure the mark; WET: re-thread on fresh webbing, soak lock + webbing 10 min in water, hang the same load wet for 1 h, measure; COLD: re-thread on fresh webbing, 2 h in the freezer, write the freezer reading, load within 2 min of taking it out, hold 10 min, measure. The record asks for -20 C: a home freezer at -18 C or colder is ACCEPTED as the -20 C condition (ASSUMPTION FREEZER; dry ice in a cool box reaches -20 C if the owner wants the exact figure)	slip < 5 mm, 5 of 5, in each of dry / wet / cold at 432 N (covers the W 338 N proof); a fail in any condition: tell the pack-designer (record fallbacks: aluminium ladderlock)	test_register.csv
SEW-F	box-X fatigue on the lever rig: 25 mm box-X loops, 1e5 cycles 44 -> 442 N (R 0.1); HE hinge eyes 62 -> 625 N on the printed fairlead (BL-044)	box-X fatigue (ASSUMPTION SEW_FAT 0.40 x box strength): IM end loops (S1), US eyes (S6), belt tabs (H1), HE hinge-eye box on the printed fairlead (BL-044, review M-3)	loop over a 5 mm pin on the lever (in place of a coupon), the free end in the bar clamp; set the crank so the cell swings 11 -> 110 N (4:1 lever = 44 -> 442 N on the loop); 1 Hz to 100,000 cycles (27.8 h); loupe check at 25,000 / 50,000 / 75,000; after run-out pull the loop on the G2 bench to 1080 N (the MT-1 US proof), 1 min. HE (BL-044; mounting BL-048 m-6, G4 rig sketch): the printed fairlead + spacer (the PT-3 fixture, the hardened steel M5 x 40 dowel (DIN 6325 m6, HRC 58-62; NOT a 7075 frame pin)) bolted to the lever rig's FIXED base plate at the coupon station, in place of the coupon block, stay segment upright under the lever's clamp; the B-05 strip in the lever's moving bar clamp (the fixture grip never holds the fabric), the pull straight down the spacer face as PT-3; cell 16 -> 156 N (62 -> 625 N on the eye), 1 Hz to 100,000 cycles, loupe on the box TOP row; after run-out, the PT-3 proof 1530 N 60 s. PIN CHECK (BL-048 M-C): roll the dowel on the bench plate before and after the test: it rolls freely with no visible wobble and a 0.05 mm feeler does not pass under its middle; a bent pin voids the specimen. PRINT (BL-048 m-6): fixtures print as the H1 frame: PAHT-CF, dried, the H1 slicer settings (print/frame_h1/README.md), annealed as H1.	loops: 3 of 3 run-outs to 1e5 at 442 N, no broken stitch, then hold 1080 N 1 min; HE: 3 of 3 run-outs to 1e5 at 625 N with no broken stitch in the box top row, then hold 1530 N 60 s (a 2 of 3 result in either set: run 3 more of that set, pass with 5 of 6)	test_register.csv

ID	Test	Gates	Steps	Pass / fail	Source
T9	loaded CG (+ 15 in, 30 kg meat check); CORRIE F rev 3 items in rows T9-R3a..d (BL-038 split)	loaded CG on the first article (rev C basis); its rev 3 items are the T9-R3a..d rows below (BL-038 split)	68 kg fill and meat on IM1-IM3 at 15/19/23 in; hang from haul loop, then from a hip pin, plumb both lines; add 30 kg on IM1-IM3 at 15 in; then rows T9-R3a (port guys, S kit CG), T9-R3c (crown strap path) in the same session; T9-R3b (cinch after a walk) and T9-R3d (lid) after T1-T4	bag CG <= 205 mm at z <= 590; meat <= 190; tall <= 185; rev 3 items: rows T9-R3a..d	test_register.csv
T9-R3a	T9 rev 3 (a): port guys hold the S-kit top, meat CG at 15 in	rev 3 (a): B-27 port guys hold the S-kit top, CG at the S kit (BL-030)	top packed full above the guys; 30 kg sand meat on IM1-IM3, fill to 68 kg; hang from the haul loop and plumb a line; hang from a pin through the hip pivot and plumb a second line; the CG is where the lines cross; measure its distance aft of the stay plane	meat CG <= 190 mm aft (record against 179.3 mm)	test_register.csv
T9-R3b	T9 rev 3 (b): crown strap cinch force and retention (CROWN_PRE_N, MU_LID)	rev 3 (b): B-28 crown strap cinch force and retention	hook the spring scale on each B-28 tail; pull to 8.2 kg (80 N) and lock; walk 30 min loaded (only after T1-T4 pass); read the tension again (pull the tail until it just moves); set the pack down hard from 0.5 m on its base; read again; check the lid is still seated	after the walk >= 40 N, or the lid still seated on the full top (report both values)	test_register.csv
T9-R3c	T9 rev 3 (c): crown strap free length (ROLL-PATH, TORSO-LIN)	rev 3 (c): B-28 crown strap free length per crown kit (cut = longest path + 40 mm spare + 2 grips, BL-039)	cinch both crown straps; measure the strap path from the bar tab to the ladder lock along the strap with a tape (both straps); repeat at the L kit 6-day setting	S kit full: path <= 968.1 mm (cut 1068.1 - 2 x 50 mm grip tail; sim 928.1 mm); L kit 6-day: record against the sim's 438.8 mm	test_register.csv
T9-R3d	T9 rev 3 (d): sewn lid mass and seat (LID_SLIDER5_G, LID_THREAD, LID-SEAT)	rev 3 (d): sewn B-29 lid mass and seat	weigh the lid alone 3 times; photograph it seated on the packed 6-day bag and the full bag, from the rear 3/4 and the side	lid 47.8 g +/- 5 g (ledger); lid edge on the hinge-band seam, camo caps over the blades	test_register.csv
FS-O5	ULTRA 200X 50 mm strip tensile (C-5 / O-5)	alternate fabric gate: ULTRA 200X 50 mm strip (floor / frame side)	clamp each strip in the 50 mm bar grips, 200 mm gauge; pull slowly to break; record the peak	mean >= 1840 N/50 mm, minimum >= 1195 N/50 mm	test_register.csv
FIT-US	US fit on P1 with the bag: clevis bolt heads clear of the bag at every crown kit end; US straps on their lanes; US2 eye over the clevis seam	BL-050 B: US straps on their lanes and the US2 clevis bolt heads on P1 with the first bag (colour transfer at kits S / M / L; the free drapes reach the heads: ASSUMPTION HEAD-PROUD)	(1) colour each clevis bolt head with lipstick or chalk paste; (2) mount the bag; 30 kg sand meat on IM1-IM3, cinch the US straps; set the crown to 15 in (kit S shortest), run the T5 crank 200 cycles (or walk 5 min if T1-T4 have passed); (3) take the bag off and look for colour on B-01, B-03 and the B-18 / floor corner; photo; (4) repeat at 19 in and 23 in (kits M, L); (5) bag-off cradle: rig the US straps round a 20 kg sand bag as a quarter, check the US2 strap and its tail clear the heads; (6) slide the US2 eye across the clevis seam in the throat floor under load (SEAM-FLOOR)	no colour transfer on the bag at any crown setting; the strap and its tail clear the heads in the cradle; the US2 eye slides across the seam without catching; FAIL: photo to the pack-designer and the structural-fea-engineer (frame head change)	test_register.csv
B-28 slip	B-28 crown strap on its heavy ladder lock: covered by LL-H at 432 N (BL-031 dropped the 20 mm gate; 25 mm strap on ladder_lock_25mm_heavy)	release of this package	B-28 crown strap on its heavy ladder lock: covered by LL-H at 432 N (BL-031 dropped the 20 mm gate; 25 mm strap on ladder_lock_25mm_heavy)	LL-H pass	corrie_f_body.md 15.6 / 16
PORT-1	B-27 port mock-up: hose from above 1 min with the guy pulled at 45 deg; 100 N guy pull through the lap; gloved use of the guy ladder lock where it is sewn: on its LLT tab at B-22, INSIDE the hinge band (DECISION GUY-LOCK-IN, BL-040 M-5), reached through the open roll top	release of this package	B-27 port mock-up: hose from above 1 min with the guy pulled at 45 deg; 100 N guy pull through the lap; gloved use of the guy ladder lock where it is sewn: on its LLT tab at B-22, INSIDE the hinge band (DECISION GUY-LOCK-IN, BL-040 M-5), reached through the open roll top	no water inside; lap stays closed at 100 N; 5 of 5 gloved adjustments at B-22	corrie_f_body.md 13.9, 13.5
GT-1	capstan pull: 5 kg on the guy over the packed bag's corner	release of this package	capstan pull: 5 kg on the guy over the packed bag's corner	record MU_WEB_FAB (0.20-0.40 band)	corrie_f_body.md 13.9
ALT-1	alternate set: owner sees an ULTRAGrid swatch	release of this package	alternate set: owner sees an ULTRAGrid swatch	owner OK in writing	part A README section 1
ALT-2	alternate set: PT-3 on 200X + 50 mm strip test	release of this package	alternate set: PT-3 on 200X + 50 mm strip test	>= 36.8 N/mm (1840 N / 50 mm mean, FS-O5)	part A README; register FS-O5
ALT-3	alternate set: weigh the 70D stock	release of this package	alternate set: weigh the 70D stock	gsm within +/- 5 % of nylon_ripstop_70d record	part A README
US-BS	US backstop: cam on 25 mm webbing with the stainless tri-glide threaded behind it; open the cam (forced slip) and load the strap to the US proof 1080 N (MT-1 basis, S5-S7 720 N x 1.5)	release of this package	US backstop: cam on 25 mm webbing with the stainless tri-glide threaded behind it; open the cam (forced slip) and load the strap to the US proof 1080 N (MT-1 basis, S5-S7 720 N x 1.5)	tri-glide stops the strap within 10 mm and holds 1080 N for 1 min, 3 of 3 (ASSUMPTION US-BS)	triglide_25mm record note; DECISIONS 2026-11-13

ID	Test	Gates	Steps	Pass / fail	Source
FRAY-1	wash-swatch fray test of the un-bound SSa2 seam (3 wash cycles, R-12)	release of this package	wash-swatch fray test of the un-bound SSa2 seam (3 wash cycles, R-12)	no allowance frayed past the 2nd row; else bind (ASSUMPTION NO-BIND)	RD-17

Defect list

Defect	Where	Cause	Prevent / fix	Class
Skipped stitches in a box-X	load points	needle too small / wrong point, flagging on Hypalon	Nm 140 R, tex 135, walking foot pressure; re-sew the whole box offset 1 mm	critical
Box-X short (stitch count under the table)	load points	early stop, wrong program	count against the load table; re-sew	critical
Hardware threaded backwards (ladder lock, cam)	straps	assembly	QC-11 hand pull; SK-7	critical
Load tab on the wrong face (B-17 out, B-18 / B-22 in)	B-01, B-02h	face mix-up	placement labels say INSIDE / OUTSIDE; check before sewing	critical
Needle-cut Hypalon / laminate (perforation line)	B-17, B-18, B-22, B-01	cutting point needle or SPI too high	R point only; 8 SPI max on Hypalon	critical
Swell puckered or twisted in its window	B-01	window not notch-matched	unpick, re-clip at the corners	major
B-27 port gapes (lap < 20 mm)	B-01	facing sewn on the wrong line	re-sew on the drawn stitch line	major
Door zip waves / snakes at the corners	door	zip stretched while sewing	basting tape, corner notches, ease into the corner	major
Seam allowance frayed past the 2nd row	body seams	straight-knife cut nylon	laser / hot-knife cut or bind (FRAY-1)	major
Notch mismatch > 2 mm / seam end mismatch > 3 mm	box seams, collar ring	pieces stretched	re-clip notch to notch, walking foot	major
Mesh pocket over-stretched	FP	mesh pulled while sewing	baste the pleats first, sew with the mesh on top	minor
Thread tension loops on the underside	any	tension / bobbin	QC-06 swatch at each set-up	minor
Burn marks / hard beads on webbing ends	webbing	hot knife too hot / slow	J8 stop, steady cut	minor
Camo piece cut against the print direction	B-02u, SH-*	lay error	ONE WAY lay rule	minor
Hook-and-loop or a non-coil zip fitted	any	substitution	BOM check (owner: quiet bag)	critical

9. Packaging

1. Bag: collar unrolled, door zipped, straps coiled inside the bag (keepers on), folded flat to about 450 x 350 mm.
2. Base kit inside the bag: IM1-IM3, US1/US2, W1-W3 fitted; guys fitted to B-22 with the toggle end coiled.
3. As-hunted kit bag (poly bag 1, labelled): B-29 lid with the B-28 straps for the ordered crown kit (S / M / L), scope pod.
4. Meat-trip kit bag (poly bag 2, labelled 'MEAT TRIP KIT'): 3 WX extenders with G-hooks, 2 B-24 links with carabiners, meat liner MS.
5. Card on top: load limits (53 kg / 116 lb frame meat in float mode, RD-02), guy rule (port guys fitted for every carry, RD-23), the strap tail rule, wash cold / line dry, no hook-and-loop.
6. Outer: recycled poly bag 600 x 800 mm, 10 bags per carton; carton label: rev 3, door D2, crown kit, fabric set (primary / alternate), date, lot.

10. Costing and hours

Step	Beginner h
0 Prep and cutting	16.28
1 Pockets	6.06
2 Small parts	4.33
3 Door D2 and hinge band	5.00
4 Main bag	14.36
5 Collar and roll top	3.50
6 Crown lid B-29 (as hunted)	8.46
7 Straps and webbing	6.46
8 Add-ons (scope pod as hunted, meat liner meat trip)	3.71
9 Rain fly F-01	5.40
10 Finish, inspect, pack	1.20
Total (incl. one-off set-up)	74.8

Scenario	SAM min	Labour $	CMT incl. overhead $	Materials $	Total $
us_contract_usd_min	509	432.44	540.55	381.06	921.61
asia_contract_usd_min	509	61.05	76.31	381.06	457.37

Lid on the drafted pattern 9.53 h (booked 7.76 h). Alternate set materials $381.14. Assumptions: SAM-RATIO 0.12; LABOUR {'us_contract_usd_min': 0.85, 'asia_contract_usd_min': 0.12}; OVERHEAD 0.25; THREAD-FACTOR 2.75; WEB_WASTE 5 %; prices with no record: PRICE_ASSUMED.

11. Release delta register: rows this release resolves

RD	Status	Evidence (checked by the script)	Still open
RD-01	closed	F-03 in the rev 3 piece list: False	nothing (v3.1 set stays superseded with its STOP notice)
RD-02	restated	B-24 x2 + 2 carabiners in the meat-trip kit BOM: True; PT-6 in the release gates	user guide (RD-22) with the 53 kg limit and belly rule; PT-6
RD-03	closed	IM1 cut 990 mm in the rev 3 cut list	MT-1 / MT-2 on the sewn straps
RD-12	restated	B-17, B-18, B-22, HE, T3, LLT, TG drafted and sequenced: True; B-19 / B-20 / B-23 / T2 removed (BL-004 REC); IM3 hooks are wiregate carabiners (BL-010)	PT-1, PT-2, PT-3, PT-5, PT-6, LL-H before any load point is sewn on a bag that will carry
RD-15	closed (numbers)	BOM, yardage and mass use the current records (robic 125 gsm, ultragrid 132 gsm 1500 mm roll, ULTRA 133 gsm)	physical: swatch weights (QC-02), FS-O5 / PT-3 on 200X (alternate gate)
RD-16	closed (release)	CORRIE F rev 3 patterns (part A) + this op sequence replace SPINE step 9	first article build; CORRIE gates; pattern-read ops listed as volatile ops
RD-17	restated	IMP-2 / IMP-4 / IMP-8 in the bag: True; SH-03 absent: True; body seams un-bound (SSa2, ASSUMPTION NO-BIND)	FRAY-1 wash-swatch test; IMP-1/3/5/6 belong to the harness / belt packages
RD-18	closed	B-11 x8 + BRL x2 + BRS + G-hook in the BOM: True; B-11 placements drawn on B-03: 8 tabs (op 4.07 released)	archery / rifle loadout fit settles ASSUMPTION B11-PLACE
RD-21	closed (pointer)	this op sequence supersedes BUILD_PLAN.md step 9 for the bag; BUILD_PLAN.md points at the rev 3 release: True	BUILD_PLAN.md body still describes the SPINE bag (its banner says so)
RD-22	open	the packaging card (README Packaging) carries the load limits	the user guide itself
RD-23	restated	B-25, B-27f, B-22 inside, TG, LLT drafted and sequenced (ops 3.04-3.05, 4.03): True	T9 rev 3 at the S kit; LL-H; PORT-1; attorney review; B-22 placed at x +/-130.3 vs drawn 149 (ASSUMPTION B22-IN)
RD-24	restated	B-29 (7 piece types) + B-28 per kit + keepers sequenced; lid 9.53 h on the drafted pattern; build_metrics books the op-sequence hours (build_effort_rev3.json: 70.31 h a bag)	T9 lid weigh / photo, cinch, free lengths

12. Assumptions that govern

ID	Assumption	Settled by
NO-BIND	Body seams are SSa2 (second row over the turned allowances), not bound: the ledger books no binding (RD-17 B-1).	FRAY-1 wash-swatch test (3 cycles); if it fails, bind with 20 mm grosgrain via a binder
NO-TAPE	No taped seams: the bag is water resistant, contents in the user's dry bag.	owner decision if a waterproof bag is wanted (seam-tape machine + taped ULTRA seams)
SAM-RATIO	Factory SAM = 0.12 x beginner hours (bag_concepts.HOURS basis).	time study on the first 10-bag lot
LABOUR	Loaded CMT minute: US contract $0.85/min, Asia $0.12/min; overhead + margin 25%.	two contract quotes from this tech pack
THREAD-FACTOR	2.75 m of thread per m of 301 stitching (both threads, ends).	weigh the cones before / after the first bag
SPI-LIGHT	10 SPI on 40D-100D and mesh.	QC-05/06 swatch: no puckering, no perforation
THREAD-NEEDLE	Tex 70 / Nm 90, tex 90 / Nm 110, tex 135 / Nm 140, round point (attach_system.NEEDLE for 90 / 135).	swatches in op 0.02: no skipped stitches, no shredding
UTIL-SMALL	Small and sheet pieces nest at 70% across bags in production.	first production markers
UTIL-ALT	Alternate-set body fabrics nest at 85% in production (no 10-bag alternate marker).	alternate 10-bag markers if the alternate set passes its gates
B11-PLACE / KW-PATH / F02-CENTRE / US-EYE	Pattern placements read by this package (patterns_corrie_f_rev3, BL-034 part A): B-11 tab heights, K-W keeper heights from the W path, F-02 fore-aft on B-05, the US eye fold.	archery / rifle loadout fits; T9 W free lengths (keeper within +/-15 mm of the wrap); fly fit; MT-1 specimens
WX-LOAD	Haul extenders carry the S9 float-clamp load (225 N).	pack-designer load case; LL-H
US-BS	The stainless 25 mm tri-glide behind each US cam stops cradle slip (record: 3.75 kN break, no slip rating).	gate US-BS: cam slipped on purpose, backstop holds
TG-SS-PRICE	Stainless tri-glide USD 2.84 from the record's EUR source line, converted as the record does (ex VAT x 1.09 USD/EUR).	supplier quote
BT-STD	Slit-end bartack 10 x 2 mm, 28 stitches; standard tack 16 x 2 mm, 42 stitches (LK-1900 patterns).	QC-07
HE-ONE / HE-PIN / HE-ENV / HE-GAP / HE-CHEEK / HE-T / HE-PIN-HI	BL-040 B-1 + BL-041 X-1 / X-2 + BL-044 Z-1 / M-3 / m-1: one hinge-eye construction (HE 96 mm folded in half, both raw ends and the B-18 edge in the floor seam, B-18 rows sewn first through B-18 + B-01 only); eye UP on the z 60 pin of the BL-042 hinge fairlead (bag_attach.HINGE_EYE_ENVELOPE: pin 10.1 mm above the stay, 6.5 mm under it (BL-049 floor drop; BL-042 5.5), 20 mm throat, 10 mm cheeks); box-X 34 x 7 at +4..+11 mm, 5 mm under the cheeks at the design pin and >= 2 mm over the HE-FIT band; eye 25 mm solved on the real 3-ply neck wrap for webbing 1.5 mm (bracket 1.2-1.6 mm, thicker rejected). The box PEELS (leg B 23-50 deg off the box line): rated by PT-3 on the printed fairlead and SEW-F HE specimens.	caliper gate on the roll (ply <= 1.6 mm); HE-FIT on the BL-042 test piece (pin 22-29 mm above the seam line, box >= 2 mm under the cheeks), then PT-3 and SEW-F HE on specimens sewn as op 4.05 / SK-8
RT-MATE / RT-HEM / RT-FWD / RT-ORIGIN / W3-OVER / HOSE-LOW / BUCKLE-ENV / MK-GAP	BL-040 B-2 + BL-041 Y-2 + BL-043 M-1 / M-2 / m-9: the RT strap is sewn at the roll end (gusset centre of the hem); its buckle's female half on an RTM tab forward on the gusset (hose slot + 10 mm + half a buckle), the buckle body 10 mm under the lowest W3 edge of every crown kit (W3 rides at the crown-bar height, z 601-804); W3 is the outer layer (slacken it to open the top); the hose slot also sits under W3's sweep; muzzle tabs 10 mm clear of the strap's line to the roll end; buckle envelope 35 x 35; RT cut from the cloth-pass roll model re-run with this tab and origin.	FIT-R3 rows 5-7 (rifle + roll clip + W3, at both ends of the owner's crown kit); T9 roll-clip free length; hydration fit (HOSE-LOW); caliper on the buckle
IM-CAM-TAB / IMC-LOOP	BL-041 m-19: each IM cam sewn to an IMC dog-bone (21 x 25 box-X each end, 18 mm loops) carrying the G-hook / carabiner.	MT-1 (2 IMC tabs at 908 N); threading the bought cam and hook
B25-TAB-CENTRE / WIDE_SLEEVE	BL-041 (crown_tube_guy_loop.md): B-25 in 38 mm webbing, sleeves x 26.5-64.5, toggle tab at the sleeve centre x 45.5; the 25 mm tab pull spreads over the 38 mm sleeve.	TL-1 at 497 N on real bar stock, specimen WITH the mitre turn and run (BL-043), rigs A / B per BL-047 (BL-048)
B25-TOPO / CLAMP (BL-045)	BL-043 M-4: B-25 = tab | sleeve | 45 deg mitre under the sleeve | run necked to 25 mm | one wrap round the spine under the clamp. BL-044: on the sized clamp (crown_spine_clamp: half 12 along the bar, collar 6) and the measured pawl housing (15.62 up, +/-24.82); the run reaches the printed spine 25.02 mm fore of the bar (+45.4 mm run). Was ASSUMED (B25-CLAMP / PAWL-TOP / SPINE_CLAMP_HALF).	BL-045 clamp T6 coupon (CLAMP-TYPE); op 7.18 gate B25-FIT at kit S 15 in; TL-1 specimen
GUY-LOCK-IN	BL-040 M-5: the guy ladder lock stays on its LLT tab at B-22 inside the hinge band; the guy length is set with the roll top open (per crown kit / load), not on the move.	PORT-1 gloved adjustment at B-22
KW-OVER-FP	BL-040 B-3: haul wraps W1 / W2 cross the FP hip pockets, over the mesh (keeper loop on top).	T9 W free lengths; first loaded haul: pocket still reachable with the wraps cinched (owner)
SHOP-INCR	Prototype buy: roll goods in 0.5 yd steps at full width, webbing whole yards, HDPE one 0.25 m2 sheet.	the owner's actual order
PRICES	Items with no material record use PRICE_ASSUMED (TPU hook sleeve, rubber pull sleeve, basting tape, seam sealer, labels, packaging). Hardware and elastic use their records since BL-034.	supplier quotes
US-CHAN-LANE / US-CUT (decisions)	The B-16 (US1) / B-16b (US2) channels end at the frame's published lane x (h1_bedfit.json: +/-70 / +/-165); US1 is cut 550 mm, US2 925 mm (us_route.json strap_lengths 'now': bag-on need 320 / 771 mm, grip left 130 / 54 mm).	HE-FIT step 0 (strap on its lane, >= 10 mm feeler) and T9 / FIT-US (cinch and grip at every kit end)

ID	Assumption	Settled by
US-CHAN-EDGE	B-16b's turned ends stay >= 15 mm inside the B-03 | B-05 side seam, so the channel end and the side seam are sewn separately (drawn: patterns geo us_chan_on_b05).	lay B-16b on the cut B-05 at op 4.14 / floor op
SEAL-PU	A urethane seam sealer bead on B-05's inside face seals the B-16b tail stitch lines on the PU-coated 500D floor (silicone does not bond to PU).	WP-1 style hose test on a B-05 offcut with one sealed tail before the bag
CRADLE-SPAN	The bag-off cradle keeps meat_haul's belly on each strap's own lane span (us_route.cradle_span: US2 legs at 23 deg, 639 N at the float meat limit, against the 720 N the frame lug was checked at; belly 50 mm would give 860 N).	FIT-US (5): US2 leg angle and tension with a 20 kg quarter stand-in in the cradle